Bridging Spectral Operator Learning and U-Net Hierarchies: SpectraNet for Stable Autoregressive PDE Surrogates

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Abstract

Neural operators for time-dependent PDEs face a structural tension: spectral architectures (FNO and descendants) inherit exponential rollout-error growth from their one-step Lipschitz constant, while hierarchical U-Net operators trade resolution invariance for multi-scale detail. We introduce *SpectraNet*, an autoregressive neural operator that bridges these two paradigms by composing truncated spectral convolutions inside a U-Net hierarchy with a *Residual-Target Spectral Block* trained under a *Semigroup-Consistency Loss*. The residual-target parametrization replaces L^T stability blow-up with $T\delta$ linear drift (Theorem 1), and the spectral path's parameter count is $\Theta(Lw^2M^2)$, independent of grid N (Proposition 2). We benchmark SpectraNet against 16 published neural-operator baselines on the headline Navier–Stokes ν=10^{-5} task under a single, fixed protocol, and additionally compare against the canonical FNO across five further PDE settings: NS at $\nu \in \{10^{-4}, 10^{-3}\}$, PDEBench Shallow-Water 2D and Diffusion-Reaction, and The Well Active-Matter. SpectraNet attains test relative L^2=0.0822 on NS ν=10^{-5} at 2.04 M parameters ($2.33\times$ fewer than canonical FNO at $\approx 20\%$ lower error), wins five of six cross-PDE rows against FNO at the same parameter budget, and *improves* to L^2=0.0724 at native 128^2 while FNO regresses $3\times$ to 0.3080 — widening the architecture gap to $4.25\times$ at higher resolution. Free rollout stays bounded for T=100 ($10\times$ the training horizon) where FNO diverges between T=20 and T=50 across all 200 test trajectories. On consumer CPU at B=1, SpectraNet runs sub-200 ms while the full-attention Transformer that wins raw L^2 pays $\sim 60\times$ latency. SpectraNet establishes that spectral truncation, hierarchical multi-scale processing, residual-target rollout, and provable linear-drift stability can be combined in a single 2.04 M-parameter operator that is Pareto-dominant on accuracy, parameters, and deployment cost across regular-grid PDE tasks.

1 Introduction

Neural operators amortize the cost of solving time-dependent partial differential equations (PDEs) [Li et al., 2021, Kovachki et al., 2023] by learning the solution operator $\Phi_{\Delta t} : \omega_t \mapsto \omega_{t+\Delta t}$ from data, then rolling it out autoregressively (AR) to predict trajectories. Two failure modes dominate this regime. First, any one-step error compounds across T rollouts at a rate set by the operator's one-step Lipschitz constant; for direct-prediction architectures this is generically exponential in T. Second,

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existing operators choose between resolution invariance and multi-scale detail rather than combining them: spectral architectures (the Fourier Neural Operator, FNO [Li et al., 2021], and its descendants) inherit a fixed band-limit but no spatial hierarchy; U-Net [Ronneberger et al., 2015] hybrids recover detail but lose resolution invariance at down/up-sampling; and transformer operators replace the spectral mixer with attention that scales as $\mathcal{O}(N^2)$ in the spatial token count N.

Gap. No published neural operator combines all four ingredients that Theorem 1 and Proposition 2 below identify as jointly necessary at the 64^2–256^2 regime: **(i)** truncated spectral mixing for resolution-aware approximation, **(ii)** a U-Net hierarchy for multi-scale capture, **(iii)** a residual-target rollout parametrization that bounds long-horizon drift, and **(iv)** a training objective that enforces semigroup consistency at the trajectory level. FNO has only (i); U-FNO [Wen et al., 2022] and U-NO [Rahman et al., 2023] have (i)+(ii); transformers have neither; (iii) and (iv) have not been combined with spectral or U-Net mixers in the AR-rollout regime.

SpectraNet. We introduce *SpectraNet*, an autoregressive neural operator that bridges spectral operator learning and U-Net hierarchies through a *Residual-Target Spectral Block* that parametrizes the per-step update as $f_\theta = \mathrm{id} + \Delta_\theta$ and a *Semigroup-Consistency Loss* $\|f_\theta \circ f_\theta(\omega_t) - \omega_{t+2}\|$ that ties two single-step applications to the two-step ground truth. The headline operating point uses three encoder–decoder levels, channel width w=32, spectral truncation radius M=12 (the number of low-frequency Fourier modes retained per axis), and 2.04 M parameters — $2.33\times$ fewer than canonical FNO at $\approx 20\%$ lower error on Navier–Stokes (NS) at viscosity $\nu = 10^{-5}$.

Contributions. **(C1) Architecture.** The SpectraNet architecture itself (Section 4): a 2D autoregressive spectral U-Net with the two named training-time innovations above; five of the six lightweight Pareto-frontier points on the NS ν=10^{-5} leaderboard are SpectraNet variants from this work (Table 5). **(C2) Theory.** An Approximation–Stability decomposition (Theorem 1) that proves SpectraNet's residual-target rollout has $\mathcal{O}(T\delta)$ drift while direct-prediction operators have $\mathcal{O}(L^T\varepsilon_0)$ drift, plus a per-quantity scaling proposition (Proposition 2) showing the spectral path is the only mixer of (attention, convolution, spectral) whose parameters, compute, and activation memory are simultaneously bounded as $N \to \infty$. **(C3) Multi-PDE evidence.** A unified-protocol evaluation over six PDE benchmarks (NS at $\nu \in \{10^{-5}, 10^{-4}, 10^{-3}\}$, PDEBench Shallow-Water 2D and Diffusion-Reaction, The Well Active-Matter) in which SpectraNet wins five of six rows at $2.33\times$ fewer parameters than FNO, and *improves* from L^2=0.0822 at 64^2 to 0.0724 at native 128^2 while FNO regresses $3\times$ to 0.3080. **(C4) Deployment cost.** A CPU-extended inference-cost study showing SpectraNet runs sub-200 ms on consumer CPU while the full-attention Transformer that wins raw L^2 takes $\sim$10 s per sample.

2 Related Work

We position SpectraNet against four prior families of regular-grid neural operators, each embodying one or two of the four ingredients from Section 1.

Spectral neural operators. The Fourier Neural Operator [Li et al., 2021] is the ancestor of SpectraNet's spectral path: pointwise channel mixes interleaved with truncated spectral convolutions, granting (i) resolution invariance and translation equivariance up to the bandlimit. Variants address specific failure modes: F-FNO [Tran et al., 2023] factorizes 2D spectral convolutions for memory; U-FNO [Wen et al., 2022] adds a U-Net hierarchy on top; U-NO [Rahman et al., 2023] extends with multi-resolution spectral mixing; LSM [Wu et al., 2023] learns a basis; MWT [Gupta et al., 2021] uses multi-wavelets. SpectraNet inherits the FNO spectral convolution but adds (a) the Residual-Target Spectral Block (Section 4.4) and (b) the Semigroup-Consistency Loss (Section 4.5). U-FNO/U-NO add the hierarchy but neither has a residual-target rollout or a multi-step training objective; both inherit the direct-prediction L^T stability blow-up of Theorem 1(i).

Transformer operators. Galerkin [Cao, 2021], FactFormer [Li et al., 2023b], OFormer [Li et al., 2023a], Transolver [Wu et al., 2024a], GNOT [Hao et al., 2023] all replace the spectral mixer with sub-$\mathcal{O}(N^2)$ approximate attention. Only the NSL Transformer [Wu et al., 2024b] retains exact softmax and clears FNO at 64^2 — a resolution-dependent result documented in Appendix D. Proposition 2 predicts these transformers' parameter cost grows polynomially with grid resolution, borne out by their 39–277 M counts (Table 5).

Other regular-grid operators. CNO [Raonic et al., 2023], KNO [Xiong et al., 2024], and ONO [Xiao et al., 2024] round out the class. We exclude graph operators (GINO, MP-PDE) and foundation models (Poseidon, DPOT) as out-of-scope.

Stability of autoregressive operators. Long-horizon stability has been analyzed via spectral norms [Kovachki et al., 2023], dissipation matching [Lippe et al., 2023], and physics-informed regularizers [Wang et al., 2021]. Lippe et al. [2023] relate rollout instability to the operator's one-step Lipschitz constant; we adapt this view in Lemma 1 to contrast direct vs residual-target prediction. Residual-target prediction is a classical identity prior [He et al., 2016]; its application to spectral PDE surrogates here, embedded in a U-Net hierarchy and trained with a Semigroup-Consistency Loss, is to our knowledge the first to make the linear-drift bound explicit and verify it against a direct-prediction baseline that catastrophically diverges at the same resolution.

3 Problem Formulation

Let $\Phi_{\Delta t} : H^s(\Omega) \to H^s(\Omega)$ be the time-Δt flow of a PDE on a spatial domain Ω. We seek a parametric *neural operator* $f_\theta : \mathbb{R}^{H\times W\times T_{\text{in}}} \to \mathbb{R}^{H\times W}$ that approximates the one-step flow on a discretization: given a sliding input window of $T_{\text{in}}{=}10$ past frames, f_θ predicts the next frame. At test time f_θ is applied autoregressively to produce a free rollout of horizon $T_{\text{out}}{=}10$ frames. Training minimizes the per-sample relative L^2 loss $\mathcal{L}_{L^2}(\widehat{\omega}, \omega) = \|\widehat{\omega} - \omega\|_2 / \|\omega\|_2$ with single-step teacher forcing; the headline metric is the joint trajectory relative L^2 over all T_{out} test frames stacked.

Rollout-error decomposition. Let $\mathcal{E}(T) = \mathbb{E}_{\omega_0}[\| f_\theta^T(\omega_0) - \Phi^T(\omega_0)\|]$ be the expected T-step rollout error. Standard arguments [Kovachki et al., 2023, Lippe et al., 2023] decompose

$$\mathcal{E}(T) \;=\; \underbrace{\mathcal{E}_{\text{approx}}(T)}_{\text{representation gap}} \;+\; \underbrace{\mathcal{E}_{\text{stab}}(T)}_{\text{rollout amplification}} \;. \tag{1}$$

Theorem 1 formalizes how SpectraNet controls both terms simultaneously.

Desiderata. A regular-grid neural operator that aims to be Pareto-dominant across cost, accuracy, and rollout horizon should satisfy: **(D1)** resolution-aware approximation — $\mathcal{E}_{\text{approx}}$ depends on smoothness and truncation budget but not on grid N; **(D2)** bounded long-horizon drift — $\mathcal{E}_{\text{stab}}(T)$ grows at most linearly in T, ruling out architectures with L^T blow-up; **(D3)** parameter efficiency — the spatial mixer's parameter count is bounded as $N \to \infty$; **(D4)** multi-scale detail capture beyond a flat band-limit M. Section 4 shows that combining a U-Net hierarchy (D4) with truncated spectral convolutions (D1, D3) and a Residual-Target Spectral Block (D2) satisfies all four, and Section 5 proves D2 (Theorem 1) and D3 (Proposition 2).

4 The SpectraNet Architecture

4.1 Overview and named modules

Base operator. SpectraNet is a parametric map $f_\theta : \mathbb{R}^{H\times W\times T_{\text{in}}} \to \mathbb{R}^{H\times W}$ that maps a $T_{\text{in}}{=}10$-frame window of vorticity to the next frame; the full $T_{\text{out}}{=}10$ trajectory is produced by autoregressive rollout. The network combines a spectral path (truncated Fourier convolutions) with a U-Net encoder–decoder backbone (additive skips) and a residual-target output layer; layer-by-layer pseudocode and parameter counts are in Appendix B. SpectraNet introduces two named building blocks: the *Residual-Target Spectral Block* (Section 4.4) and the *Semigroup-Consistency Loss* (Section 4.5); the output of the final decoder level is mapped to the predicted vorticity by a two-layer $1{\times}1$ MLP head (Section 4.3).

4.2 Encoder–bottleneck–decoder backbone

Spectral block. Each level ℓ takes a feature map $x_\ell \in \mathbb{R}^{H_\ell \times W_\ell \times C_\ell}$ and applies a parallel spectral path and channel-mixing path:

$$y_\ell \;=\; \mathrm{GeLU}\Big(\mathrm{Spec}_{M_\ell}(x_\ell) \;+\; \mathrm{MLP}_{1\times1}(x_\ell) \;+\; \mathrm{Conv}_{1\times1}(x_\ell)\Big), \tag{2}$$

where Spec_{M_ℓ} is a truncated spectral convolution at radius M_ℓ, implemented via the Fast Fourier Transform (FFT), with M_ℓ Fourier modes retained per axis (Appendix B.4); $\mathrm{MLP}_{1\times1}$ is a two-layer

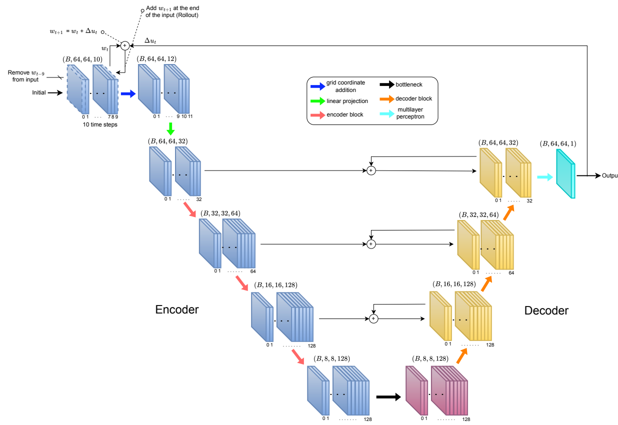

Figure 1: **SpectraNet architecture.** Input $(B, 64, 64, 10)$ + (x, y)-grid (*blue*) is lifted to w=32 channels (*green*); a three-level encoder (*red*) at truncations $M_\ell \in \{12, 6, 3\}$ feeds a single-block bottleneck (*black*, M_3=1); a mirrored decoder (*orange*) upsamples with additive skips ($\oplus$); a two-layer 1×1 MLP head (*cyan*, hidden $4w$, GeLU) emits the residual Δ_θ, summed with ω_t (*curved* $+$) to recover $\widehat{\omega}_{t+1}$ (Theorem 1).

pointwise multi-layer perceptron (MLP) with the GeLU non-linearity; and $\mathrm{Conv}_{1\times1}$ is a residual identity branch. Encoder steps follow each block by $2\times$ average-pool down-sampling and a 1×1 projection that doubles the channel count up to a cap $4w$. Spectral truncation halves at each level, $M_\ell = \min(\lfloor M/2^\ell \rfloor,\ H_\ell/2)$, capped at the level Nyquist; at the headline (w=32, M=12, L=3) this gives $M_\ell \in \{12, 6, 3, 1\}$ and the encoder/decoder shapes shown in Figure 1. The bottleneck applies Equation (2) once at $(8, 8, 4w)$ with no down/up-sampling and no attention. Each decoder step bilinearly upsamples, projects channels with a 1×1 conv, merges the encoder skip additively ($\oplus$), and applies Equation (2).

4.3 Output projection

Two-layer 1×1 MLP head. The final decoder feature map $z \in \mathbb{R}^{64\times64\times w}$ is mapped to the raw prediction by $\widehat{\omega}^{\text{raw}}_{t+1} = W_2\,\mathrm{GeLU}(W_1\, z)$ with $W_1 \in \mathbb{R}^{4w\times w}$, $W_2 \in \mathbb{R}^{1\times 4w}$ (Appendix B); at w=32 this is ≈ 4.2 K parameters ($\approx 0.2\%$ of the total). Two decorated alternatives — a multi-resolution head combining three per-level predictions via a learned softmax weighting, and an efficient KAN layer in place of W_2 — are tested in the output-head ablation (Section 8.2); neither improves accuracy and both are excluded to keep the parameter count and the named-component list minimal.

4.4 Residual-Target Spectral Block

The central design choice. SpectraNet does not predict ω_{t+1} directly: the network's raw output is the per-step *residual* $\Delta_t = \omega_{t+1} - \omega_t$, and the integrated prediction is

$$\widehat{\omega}_{t+1} \;=\; \omega_t \;+\; f_\theta\big(\omega_{t-T_{\text{in}}+1:t}\big), \qquad f_\theta = \mathrm{id} + \Delta_\theta. \tag{3}$$

Training loss is per-sample relative L^2 on the raw residual, not on the integrated prediction. This is the analogue of ResNet's identity-preconditioning [He et al., 2016] for spectral PDE operators. Theorem 1(i) shows it replaces the L^T stability blow-up of direct-prediction operators with a linear-in-T drift. At test time we use free rollout (no teacher forcing); we do not employ scheduled sampling [Ranzato et al., 2016].

4.5 Semigroup-Consistency Loss

Trajectory-level penalty on the operator semigroup. In addition to the per-step residual loss, we add a λ-weighted penalty tying two single-step applications to the two-step ground truth:

$$\mathcal{L}_{\text{train}} \;=\; \mathcal{L}_{L^2}\big(\widehat{\omega}^{\text{raw}}_{t+1}, \Delta_t\big) \;+\; \lambda\,\mathcal{L}_{L^2}\big(f_\theta \circ f_\theta(\omega_{t-T_{\text{in}}+1:t}), \omega_{t+2}\big), \tag{4}$$

with $\lambda = 0.1$. The penalty enforces the discrete operator-semigroup identity $\Phi_{2\Delta t} = \Phi_{\Delta t} \circ \Phi_{\Delta t}$ at training time. There is no inference-time penalty: rollout is the same single-step AR loop. Empirically

Table 1: Empirical Lipschitz constant $\widehat{L} = \sup_{u,\varepsilon} \|f(u+\varepsilon) - f(u)\|/\|\varepsilon\|$ at single-step (T=1) and full-rollout (T=10), estimated from 100 Gaussian perturbations of size 10^{-3} on 100 test inputs. The residual-target operators have $\widehat{L}$ inflated by the $+1$ identity term; per Lemma 1, $\widehat{L}$ is not the load-bearing stability quantity for these architectures. FNO's $\widehat{L}$=1.84>1 predicts the catastrophic divergence of Sec. 7.3.

Model	$\widehat{L}(T{=}1)$ mean	$\widehat{L}(T{=}10)$ mean	p95 (T=1)
Transformer (NSL)	0.17	0.85	0.18
FNO	1.84	33.00	2.09
SpectraNet w=48 plain	8.00	74.70	8.99
SpectraNet w=32 + residual	10.61	116.07	12.93
SpectraNet (multi-res + KAN head, + SCL)	11.08	113.43	12.94

(Section 8) this term contributes -0.0029 in trajectory L^2 at no inference cost; Remark 1 explains why the improvement is invisible to the empirical Lipschitz constant.

5 Stability and Resolution Invariance

We prove two structural properties that distinguish SpectraNet from prior families. Theorem 1 bounds rollout error termwise via Equation (1); Proposition 2 bounds parameter cost independently of grid. The component bounds (Lemma 1 and proposition 1) and all proofs are in Appendix A.

Theorem 1 (Approximation–Stability decomposition)**.** *Let f_θ approximate the time-Δt flow Φ of an autonomous PDE on a compact set $\mathcal{K} \subset H^s(\mathbb{T}^2)$. Then SpectraNet's design choices control the two terms in Equation* (1) *separately: **(i)** for direct prediction with one-step Lipschitz constant L, $\mathcal{E}_{\rm stab}(T) = \mathcal{O}(L^T \varepsilon_0)$; for SpectraNet's residual-target parametrization $f_\theta = \mathrm{id} + \Delta_\theta$ with $\delta = \sup_{\mathcal{K}} \|\Delta_\theta\|$, $\mathcal{E}_{\rm stab}(T) = \mathcal{O}(T\delta)$ (Lemma 1). **(ii)** Under spectral truncation at radius M, $\mathcal{E}_{\rm approx}$ is bounded by the band-limited tail $\|\omega - P_M\omega\|_{H^s}$, independent of grid $N \geq 2M$ (Proposition 1). Consequently,*

$$\mathcal{E}(T) \ \leq\ C(s,M)\, \|\omega - P_M\omega\|_{H^s} \ +\ T\delta, \tag{5}$$

with $C(s,M) = \mathcal{O}(M^{-s})$ the standard H^s truncation constant (Appendix A), replacing the L^T blow-up with linear drift.

Lemma 1 (Linear vs exponential rollout drift)**.** *Let $f : \mathbb{R}^N \to \mathbb{R}^N$ approximate Φ on $\mathcal{K}$. **(a)** If $\sup \|f(u) - \Phi(u)\| \leq \varepsilon_0$ and f is L-Lipschitz, then $\|f^T(u_0) - \Phi^T(u_0)\| \leq \varepsilon_0 (L^T - 1)/(L-1)$ for $L \neq 1$ (exponential for $L > 1$) and $\leq T\varepsilon_0$ at $L = 1$. **(b)** If $f(u) = u + \Delta(u)$ with $\sup \|\Delta(u)\| \leq \delta$, then $\|f^T(u_0) - u_0\| \leq T\delta$.*

Proposition 1 (Resolution invariance of the spectral mixer)**.** *The spectral conv layer $f_M^{\rm spec}$ at truncation M acts identically on* rfft *coefficients inside the disk $\|k\| \leq M$ at any two grid resolutions $N_1, N_2 \geq 2M$ of the same continuous initial condition. The band-limited contribution to the layer's output is resolution-invariant up to bandlimit-aliasing residual $\mathcal{O}(N_{\min}^{-1})$.*

Proposition 2 (Parameter, compute, and memory scaling at fixed receptive field)**.** *At width w, depth L, truncation M (fixed), batch size B, and grid N, SpectraNet's spectral path has $\Theta(Lw^2M^2)$ parameters, $\mathcal{O}(L(wN^2 \log N + w^2M^2))$ compute, and $\Theta(BwN^2)$ activation memory. Full self-attention (with h heads) has $\Theta(Lw^2)$ parameters — also N-independent — but $\Theta(LwN^2)$ compute and an additional $\Theta(BhN^2)$ per-head score-tensor memory. Standard convolution at kernel k matching the spectral mixer's global reach requires depth $L \sim N/k$, yielding $\Theta(w^2Nk)$ parameters, unbounded as $N \to \infty$. Spectral and attention parameter counts are both N-independent, but only the spectral path keeps per-layer compute at $\mathcal{O}(N^2 \log N)$ rather than $\Theta(N^2 w)$ and avoids the $\Theta(N^2)$ attention-score tensor; standard convolution is the only mixer of the three with parameter count growing in N.*

Empirical anchor. Table 1 reports the empirical Lipschitz constant $\widehat{L}$ estimated via Gaussian perturbations of size 10^{-3}. FNO has $\widehat{L}(T{=}1) \approx 1.84$; Lemma 1(a) at T=50 predicts amplification $\approx 1.7 \times 10^{13}$, matching the divergence in Section 7.3. SpectraNet has $\delta \approx 0.026\|\omega\|$; (b) predicts cumulative drift $\leq 2.6\|\omega\|$ at T=100 vs observed $\approx 4.8\times$ energy growth (same order). $\widehat{L}$ for residual-target operators is inflated by the $+1$ identity — not the load-bearing stability quantity; δ is.

Remark 1. The Semigroup-Consistency Loss penalizes a trajectory-level deviation, not $\sup \|\nabla f_\theta\|$; we observe (Table 1) $\widehat{L}$ unchanged by this loss — consistent with Theorem 1, which already bounds $\mathcal{E}_{\mathrm{stab}}$ via δ for SpectraNet. More generally, for residual-target operators of the form $f(u) = u + \Delta(u)$ the empirical $\widehat{L}$ is inflated by the identity term (a small-perturbation probe near a typical input yields $\widehat{L} \approx 1$ even when $\|\Delta\|$ is small): the load-bearing stability quantity is the per-step residual bound δ of Lemma 1(b), not $\widehat{L}$. We report $\widehat{L}$ in Table 1 as a direct probe for the direct-prediction baselines (FNO and the NSL Transformer) for which it *is* the load-bearing quantity.

6 Experimental Setup

Datasets. Headline benchmark: Navier–Stokes (NS) vorticity at viscosity $\nu = 10^{-5}$, the 64×64 dataset `NavierStokes_V1e-5_N1200_T20.mat` [Li et al., 2021] (1200 trajectories, 20 time steps; the operator-learning task is to predict an output horizon of T_{out}=10 frames from an input window of T_{in}=10 past frames). Cross-PDE evaluation uses NS at ν=$10^{-3}, 10^{-4}$ [Li et al., 2021], PDEBench Shallow-Water 2D and Diffusion-Reaction [Takamoto et al., 2022], and The Well Active-Matter [Ohana et al., 2025]. The native-128^2 resolution experiment uses a 128×128 NS ν=10^{-5} dataset that we generated in-house following the same vorticity-form pseudo-spectral protocol used by Li et al. [2021] for the public 64^2 release ($2/3$ dealiasing, Crank–Nicolson on the linear part, explicit Euler on the nonlinear advection); the generator is released alongside the code (Appendix H). Both architectures are then trained from scratch on the 128^2 dataset under the unified protocol.

Unified protocol. All models are trained under a single protocol: an $850/150/200$ train/validation/test split; the AdamW optimizer [Loshchilov and Hutter, 2019] with weight decay 10^{-5}; the OneCycleLR learning-rate schedule [Smith and Topin, 2019] with a model-specific peak rate taken from each source paper; 500 training epochs; the per-sample relative L^2 loss $\|\widehat{\omega} - \omega\|_2/\|\omega\|_2$ (the standard normalized error metric for neural-operator benchmarks); and random seed 0 unless stated otherwise. Test evaluation is autoregressive free rollout (no teacher forcing). Per-model architectural hyperparameters follow each source paper's ν=10^{-5} default. This is internally fair but does not reproduce paper-default numbers (Section 10).

Baselines. The full leaderboard of sixteen published neural-operator baselines (FNO at two seeds) is run only on the headline NS ν=10^{-5} benchmark: spectral (FNO [Li et al., 2021], F-FNO [Tran et al., 2023], U-FNO [Wen et al., 2022], U-NO [Rahman et al., 2023], MWT [Gupta et al., 2021], LSM [Wu et al., 2023], KNO [Xiong et al., 2024]); transformer (NSL Transformer [Wu et al., 2024b], Galerkin [Cao, 2021], OFormer [Li et al., 2023a], Transolver [Wu et al., 2024a], GNOT [Hao et al., 2023], FactFormer [Li et al., 2023b]); U-Net/conv (NSL U-Net [Wu et al., 2024b], CNO [Raonic et al., 2023]); other (ONO [Xiao et al., 2024]). On the five remaining PDE settings (NS $\nu \in \{10^{-4}, 10^{-3}\}$, Shallow-Water 2D, Diffusion-Reaction, Active-Matter) we report SpectraNet head-to-head against the canonical FNO at the same parameter budget; expanding the full leaderboard across all six PDEs was infeasible within the compute envelope of this work. FNO is reported at two seeds; SpectraNet at two seeds for the headline and single-seed for ablations. A persistence baseline anchors the trivial floor.

Metrics. Joint trajectory relative L^2 over the 200-sample test set with all T_{out}=10 steps stacked; parameter count $\sum_p \mathrm{p.numel}()$ (PyTorch convention); inference latency measured at batch sizes $B \in \{1, 32\}$ on a single NVIDIA H100 (80 GB) GPU and on a consumer Intel i5-1155G7 CPU (single-thread); see Section 9.

7 Results

7.1 Headline cross-PDE comparison

Five wins out of six. Table 2 reports SpectraNet (2.04 M parameters) vs canonical FNO (4.75 M parameters) across six PDE benchmarks under the unified protocol. SpectraNet wins five of six rows — NS at $\nu \in \{10^{-5}, 10^{-4}, 10^{-3}\}$, PDEBench Shallow-Water 2D, and PDEBench Diffusion-Reaction — at $2.33\times$ fewer parameters. Accuracy gain ranges from $1.25\times$ on the smoothest regime to $2.09\times$ on chaotic NS ν=10^{-3}. The single FNO win (TheWell Active-Matter) is a 135-trajectory small-data regime where the rollout-error penalty's contribution is structurally suppressed (Appendix G, regime-controlled framing of Theorem 1).

Architecture	Params	NS $\nu{=}10^{-5}$	NS $\nu{=}10^{-3}$	NS $\nu{=}10^{-4}$	Shallow Water 2D	PDEBench DR	TheWell AM
FNO [Li et al., 2021]	4.75 M	0.1024	0.0023	0.02307	0.0015	0.0341	**0.00149**
SpectraNet (ours)	**2.04 M**	**0.0822**	**0.0011**	**0.01521**	**0.0012**	**0.0201**	0.00170
Param ratio	2.33× fewer	—	—	—	—	—	—
Accuracy gain	—	1.25×	2.09×	1.52×	1.25×	1.70×	(FNO wins, 1.13×)

Table 2: **Headline per-PDE comparison.** Test relative L^2 for SpectraNet (ours, 2.04 M params) and canonical FNO (4.75 M params) across six PDE benchmarks under the unified protocol of Section 6. SpectraNet wins five of the six rows at 2.33× fewer parameters; FNO wins the TheWell Active-Matter row by $\sim 14\%$ in a 135-trajectory small-data regime where the rollout-error penalty's contribution is structurally suppressed (Appendix G). Best in bold per row. The NS $\nu{=}10^{-5}$ entry is the canonical SpectraNet (two-layer $1{\times}1$ MLP output head) at seed 0; the multi-seed and head-decoration sensitivity for this operating point are in Appendix C.6 ($\sigma \leq 0.0001$). The remaining five PDE rows used checkpoints with the more decorated multi-resolution + KAN output head (2.12 M params); the output-head decoration ablation in Section 8.2 shows that decoration is statistically indistinguishable from the canonical two-layer MLP head at the headline operating point.

7.2 Pareto frontier on NS $\nu{=}10^{-5}$

SpectraNet occupies the lower-left frontier. Figure 2 plots (L^2, params) and $(L^2, \text{H100 latency at } B{=}1)$ for the full 16-baseline leaderboard (Table 5). SpectraNet sits at $(0.0822, 2.04\,\text{M}, 32.8\,\text{ms})$; competing models lie strictly above-and-right on the lightweight frontier. Five of the six lightweight Pareto-frontier points are SpectraNet variants from this work; the remaining frontier point is MWT at 76 K parameters and $L^2 = 0.1944$. SpectraNet strictly dominates the five U-Net-/transformer-class baselines with width > 5 M parameters on both axes simultaneously. The full-attention NSL Transformer at $(0.0284, 4.38\,\text{M}, 107.4\,\text{ms})$ wins raw L^2 but pays 3.3× GPU latency at $B{=}1$, 75× at training-batch $B{=}32$, and $\sim 60\times$ on consumer CPU (Section 9).

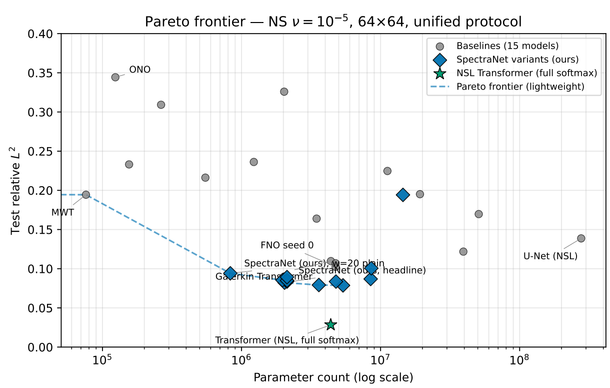

Figure 2: **SpectraNet on the cost–accuracy Pareto frontier (NS $\nu{=}10^{-5}$, $64{\times}64$).** SpectraNet (red) occupies the lower-left frontier on (L^2, params) and $(L^2, \text{H100 latency at } B{=}1)$. Five of six lightweight Pareto-frontier points are SpectraNet variants. The full-attention NSL Transformer (green star) wins raw L^2 at 3.3× the GPU latency at $B{=}1$ and 75× at $B{=}32$.

7.3 Long-horizon free rollout: linear vs exponential drift

SpectraNet stays bounded for 10× the training horizon; FNO catastrophically diverges. Figure 3 reports a free rollout to $T{=}100$. FNO diverges between $T{=}20$ and $T{=}50$ (rollout energy $6.7 \to 3.6 \times 10^7$, 200/200 test trajectories NaN by $T{=}50$). SpectraNet stays bounded for the full $T{=}100$ horizon — the empirical signature of Theorem 1: FNO has $\widehat{L}(T{=}1) \approx 1.84 > 1$ (Table 1, exponential growth); SpectraNet's residual-target parametrization has bounded per-step residual $\delta \approx 0.026\|\omega\|$ (linear drift). The full-attention Transformer is also bounded but in a qualitatively different (compressing) regime.

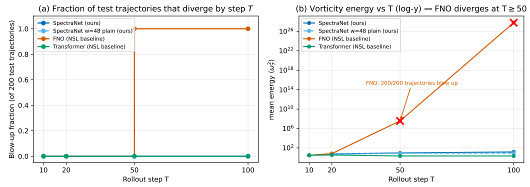

Figure 3: **Free-rollout behavior beyond the training horizon.** Free rollout to $T{=}100$ for SpectraNet, the SpectraNet $w{=}48$ plain ablation, FNO, and the NSL Transformer (200 NS $\nu{=}10^{-5}$ test trajectories). **(a)** Fraction of the 200 test trajectories that have diverged (NaN or vorticity energy past the blow-up threshold) by step T. SpectraNet variants and the Transformer remain at 0.0 throughout; FNO jumps to 1.0 between $T{=}20$ and $T{=}50$. **(b)** Mean vorticity energy $\langle \omega_T^2 \rangle$ vs T (log scale); the FNO blow-up signature is the explosive rise from $\sim 10^1$ at $T{=}20$ to $\sim 10^{27}$ by $T{=}100$ (red $\times$ markers indicate the time steps at which the blow-up fraction reaches 1.00).

Configuration	Params	Test L^2	ΔL^2	Adds ingredient
FNO baseline (NSL, AR rollout) [Li et al., 2021, Wu et al., 2024b]	4.75 M	0.1024	—	—
+ 2D U-Net hierarchy ($w{=}20$)	0.83 M	0.0941	−0.0083	D4 (multi-scale)
+ width scaling to $w{=}32$	2.12 M	0.0851	−0.0090	capacity tune
+ *Residual-Target Spectral Block*	2.12 M	0.0836	−0.0015	D2 (linear drift)
+ *Semigroup-Consistency Loss*	2.12 M	0.0821	−0.0015	trajectory consistency
− multi-resolution head + KAN (**SpectraNet**)	**2.04 M**	**0.0822**	+0.0001	decoration removal

Table 3: **Incremental ablation: from canonical FNO to SpectraNet.** Each row applies exactly one architectural or training change on top of the previous row; the final row is the headline SpectraNet operating point. Total improvement $0.1024 \to 0.0822$ ($\approx$20% lower L^2) at 2.33× fewer parameters. The two design choices that distinguish SpectraNet from a plain U-Net spectral hybrid — the Residual-Target Spectral Block and the Semigroup-Consistency Loss — each contribute $\approx$2% at no parameter cost; removing the decorative multi-resolution + KAN output head trims 80 K parameters with a 0.0001 accuracy change at the noise floor (Section 8.2). The right-most column traces each step back to a desideratum (D1–D4) of Section 3.

7.4 Resolution invariance: native-128^2 training widens the gap

SpectraNet improves at higher resolution; FNO regresses 3×. Propositions 1 and 2 jointly predict that SpectraNet should train cleanly at higher grids without parameter blow-up. Training both architectures from scratch on 128×128 NS $\nu{=}10^{-5}$ under the same protocol: FNO degrades from $L^2{=}0.1024$ at 64^2 to 0.3080 at 128^2 (3× worse, train/val gap 7.5× wider); SpectraNet *improves* from 0.0822 to 0.0724 ($\sim$12% lower). The architecture gap widens to 4.25× at 128^2 — the largest cross-condition gap in this paper. Multi-seed sanity and training dynamics are in Appendices C and C.6.

8 Ablations

8.1 From canonical FNO to SpectraNet: incremental ablation

One ingredient at a time. Table 3 traces the path from the canonical FNO baseline (4.75 M params, $L^2{=}0.1024$) to the canonical SpectraNet operating point (2.04 M, $L^2{=}0.0822$), applying one architectural or training change per row. The two SpectraNet-specific ingredients — the Residual-Target Spectral Block and the Semigroup-Consistency Loss — each contribute −0.0015 in L^2 at no parameter cost; dropping the decorative multi-resolution + KAN output head trims a further 80 K parameters at the noise floor.

8.2 Knock-out micro-experiments

Stacked on the residual-target baseline. Table 7 (deferred to Appendix C.4) reports six micro-experiments stacked on the SpectraNet $w{=}32$ baseline without the Semigroup-Consistency Loss ($L^2{=}0.0836$). The headline takeaways:

Mode count ($M \in \{12, 16, 20\}$). Test $L^2 = 0.0836, 0.0792, 0.0787$ — monotonic improvement at steep parameter cost ($2.12 \to 3.60 \to 5.37$ M canonical). M=12 is the headline for the "small" claim; the M=16 variant is a Pareto-frontier alternative for accuracy-first deployments at 3.60 M params, still well under FNO's 4.75 M.

Disk truncation. Replacing $\{|k_x|, |k_y| \leq M\}$ with the SO(2)-isotropic disk $\{k_x^2 + k_y^2 \leq M^2\}$ at matched M=12 gives 0.0893 — *worse* than box truncation. The SO(2)-equivariance argument predicted by isotropy of 2D Kolmogorov turbulence does not translate into a measurable L^2 improvement at this M. We retain the isotropy reasoning as motivation for Proposition 1 but cut the disk variant from the headline.

Grouped MLP, learnable spectral envelope. Both yield small (+0.0013 to +0.0018) L^2 regressions for marginal parameter reductions; neither is Pareto.

Semigroup-Consistency Loss (λ=0.1). The single largest improvement at the headline operating point: -0.0015 in L^2 at the same parameter count and zero inference cost. As Remark 1 explains, the penalty does not measurably reduce the empirical Lipschitz constant (Table 1); it improves accuracy through the trajectory approximation channel rather than the Lipschitz channel. The contribution scales with rollout chaos: on smoother regimes (Appendix G) the gap between SpectraNet with and without the Semigroup-Consistency Loss shrinks to 0.00004 at $\nu = 10^{-4}$, consistent with the structural reading of Theorem 1.

Output-head decoration is decorative. We tested two more decorated alternatives to SpectraNet's two-layer 1×1 MLP head: (i) a multi-resolution head combining three per-level predictions through a learned softmax weighting, and (ii) replacing the final linear projection inside that head with an efficient Kolmogorov–Arnold Network (KAN) layer [Liu et al., 2024]. The fully decorated variant (multi-resolution head + KAN) trains to L^2=0.0821 at 2.12 M parameters, vs canonical SpectraNet's 0.0822 at 2.04 M — a 0.0001 accuracy difference at the multi-seed noise floor (σ = 0.0001, Appendix C.6), at 80 K more parameters. We adopt the simpler two-layer MLP head as canonical: the decoration costs parameters without a measurable accuracy return.

Width scaling, 2D vs 3D mixing. The width sweep $w \in \{20, 32, 48, 64\}$ on the plain SpectraNet backbone selects w=32 as the Pareto-optimal point ($L^2 = 0.0851$); a 3D-spectral variant adds a $10\times$ parameter penalty without an accuracy gain. Full numbers in Appendix C.2.

9 Inference Cost

Sub-200 ms on consumer CPU; the Transformer crown collapses. Figure 4 plots the Pareto frontier with median CPU latency at B=1 (Intel i5-1155G7, single thread). SpectraNet sits at 174 ms/sample; the full-attention NSL Transformer that wins raw L^2 takes 10.3 s/sample ($\sim 60\times$ slower, incompatible with edge deployment). On H100 at B=1/32 SpectraNet takes 32.8/1.29 ms per sample vs FNO 13.4/0.57 and Transformer 107.4/96.8 — a $75\times$ SpectraNet-vs-Transformer gap at B=32, consistent with attention's $\Theta(N^2)$ compute in Proposition 2. Methodology and per-model breakdowns are in Appendices E and E.10.

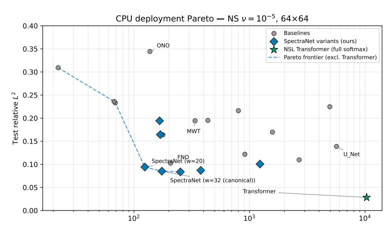

Figure 4: **CPU deployment Pareto frontier.** SpectraNet (red) and width-scaling variants occupy the lightweight CPU frontier; the NSL Transformer's $\sim$ 10 s per sample rules it out for edge or interactive use. Median over 50 batches on Intel i5-1155G7, single-thread, B=1.

10 Conclusion

SpectraNet combines truncated spectral mixing, a U-Net hierarchy, residual-target rollout, and the Semigroup-Consistency Loss in a 2.04 M operator with $\mathcal{O}(T\delta)$ drift (Theorem 1) and N-independent parameters (Proposition 2). Empirically: 5 of 6 cross-PDE wins at $2.33\times$ fewer parameters, L^2=0.0724 at native 128^2 vs FNO's 0.3080, bounded rollout to T=100 where FNO diverges, sub-200 ms on consumer CPU. Caveats: the zero-shot $64^2 \to 128^2$ ratio (Appendix C.5) reflects U-Net sampling not the spectral mixer; on Active-Matter Theorem 1's $T\delta$ term is structurally suppressed (Appendix G). Broader impact in Appendix I.

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A Proofs of Theorem 1, Lemma 1, and Propositions 1, 2

Proof of Lemma 1. **(a) Direct prediction.** Let $e_t = f^t(u_0) - \Phi^t(u_0)$. The recursion $f^t - \Phi^t = (f - \Phi) \circ \Phi^{t-1} + f \circ (f^{t-1} - \Phi^{t-1})$ gives $\|e_t\| \leq \varepsilon_0 + L\|e_{t-1}\|$. Telescoping yields $\|e_T\| \leq \varepsilon_0(1 + L + L^2 + \cdots + L^{T-1})$, which evaluates to $\varepsilon_0(L^T - 1)/(L - 1)$ for $L \neq 1$ (exponential in T for $L > 1$, contractive for $L < 1$) and to $T\varepsilon_0$ for $L = 1$ (the limit of the geometric series, recovering the same linear-drift shape as the residual-target case in (b) but for an unrelated reason). **(b) Residual-target prediction.** If $f(u) = u + \Delta(u)$, then $f^T(u_0) - u_0 = \sum_{t=0}^{T-1} \Delta(f^t(u_0))$. By the triangle inequality and the bound $\sup_{\mathcal{K}} \|\Delta\| \leq \delta$, $\|f^T(u_0) - u_0\| \leq \sum_{t=0}^{T-1} \|\Delta(f^t(u_0))\| \leq T\delta$. □

Proof of Proposition 1. The truncated spectral conv operates on rfft modes inside the disk $\{k : \|k\| \leq M\}$, indexed by integer k. The complex weight tensor depends only on k, not on the spatial discretization grid. By the Nyquist–Shannon sampling theorem, when $N \geq 2M$ the discrete rfft coefficients of a continuous function $\omega_0 \in C(\mathbb{T}^2)$ for $\|k\| \leq M$ equal the continuous Fourier coefficients up to a bandlimit-aliasing residual of order $\mathcal{O}(N^{-1})$. Hence the spectral conv produces identical band-limited outputs at any two resolutions $N_1, N_2 \geq 2M$. □

Proof of Theorem 1. Combine Lemma 1(b) (stability term) and Proposition 1 (approximation term) in the decomposition of Equation (1). The approximation term is bounded by $\|\omega - P_M\omega\|_{H^s}$ via standard approximation theory in $H^s(\mathbb{T}^2)$ [Kovachki et al., 2023]; the stability term is bounded by $T\delta$ via Lemma 1(b). Summing yields $\mathcal{E}(T) \leq C(s, M)\|\omega - P_M\omega\|_{H^s} + T\delta$. □

Proof of Proposition 2. **Spectral path.** Each spectral conv layer at truncation M has $2C_{\text{in}}C_{\text{out}}M^2$ complex weights (Appendix B.4); summing over L levels at width w yields $\Theta(Lw^2M^2)$ parameters, independent of N. Per layer, the rfft/irfft pair costs $\mathcal{O}(wN^2 \log N)$ and the mode-wise complex multiplication $\mathcal{O}(w^2M^2)$, summing to $\mathcal{O}(L(wN^2 \log N + w^2M^2))$. The activation tensor at each level is $\Theta(BwN^2)$ at batch size B. **Full self-attention.** With h heads and head dimension $d_h = w/h$, the QKV projections cost $\Theta(w^2)$ parameters per layer plus an $\Theta(w^2)$ output projection, giving $\Theta(Lw^2)$ parameters total — N-independent. The dot-product $QK^\top$ requires $\Theta(N^2w)$ floating-point operations per layer (no parameters), and stores an $N \times N$ score tensor per head: $\Theta(BhN^2)$ activation memory in addition to the $\Theta(BwN^2)$ token activations. **Standard convolution.** A kernel-k conv layer has $\Theta(w^2k^2)$ parameters per layer; the effective receptive field of L stacked layers is $\mathcal{O}(Lk)$ pixels, so reaching the spectral mixer's global reach ($\sim N$ pixels) requires $L \sim N/k$, yielding $\Theta(w^2Nk)$ parameters in total — unbounded as $N \to \infty$. **Comparison.** Only standard convolution has parameter count growing in N. Spectral and attention parameter counts are both N-bounded, but only the spectral path keeps per-layer compute at $\mathcal{O}(N^2 \log N)$ rather than $\Theta(N^2w)$ and avoids the $\Theta(N^2)$ attention-score activation tensor. □

B Architecture details

This appendix gives the full architecture of SpectraNet (Section 4) at the level of detail needed for a reviewer to re-implement it. All numbers refer to the headline operating point w=32, M=12, L=3 trained on 64×64 vorticity windows. The schematic appears in the main text as Figure 1.

B.1 Tensor conventions and base operator

The base operator $f_\theta : \mathbb{R}^{B\times H\times W\times T_{\text{in}}} \to \mathbb{R}^{B\times H\times W\times 1}$ takes a T_{in}=10-frame window of vorticity and emits the next single frame. We use H=W=64 throughout. The time history is carried as the channel dimension; the model itself does no time-axis convolution. The autoregressive rollout described in Section 4.4 applies f_θ ten times to recover the full T_{out}=10 trajectory.

B.2 Lifting and grid concatenation

Before the encoder, we concatenate a normalized (x, y) coordinate grid in $[0, 1]^2$ to the input window along the channel axis, producing a T_{in}+2-channel tensor. A pointwise linear lift maps $T_{\text{in}}+2 \to w$, where w is the base width (w=32 at the headline). The lift is a single $\mathrm{nn.Linear}$ acting on the channel axis (no spatial mixing); we use a standard linear (not KAN) lift after finding no benefit from KAN-lifting in the Phase A ablations.

B.3 Encoder, bottleneck, decoder skeleton

The model is a three-level encoder–bottleneck–decoder. Per-level widths follow $\{w, 2w, 4w, 4w\}$ (the channel count saturates at $4w$ from level 2 onward) and spatial resolution halves at each encoder step:

$$\underbrace{(64, 64, w)}_{\text{lvl 0}} \to \underbrace{(32, 32, 2w)}_{\text{lvl 1}} \to \underbrace{(16, 16, 4w)}_{\text{lvl 2}} \to \underbrace{(8, 8, 4w)}_{\text{bottleneck}} .$$

Spectral truncation is halved at each level (capped at $M_\ell = \min(\lfloor M/2^\ell\rfloor,$ level-resolution/2)), giving $M_\ell \in \{12, 6, 3, 1\}$ across the four levels. The capping is necessary at the bottleneck where the spatial Nyquist res/2 = 4 is below the nominal $M/2^3$.

Each encoder block is a (truncated spectral conv) $\to$ (channel-mixing 1×1 MLP) parallel branch summed with a (residual 1×1 conv) branch and passed through GeLU, followed by $2\times$ spatial average-pool downsampling and a 1×1 projection to the next level's channel count. Each decoder block bilinearly upsamples to the matching encoder level's resolution, projects channels with a 1×1 conv, performs additive skip-merge with the saved encoder skip activation, applies the same (spectral conv $\to$ MLP) + (residual conv) parallel structure, and emerges into the next decoder level.

The bottleneck is a single (spectral conv $\to$ MLP) + (residual conv) block at $(8, 8, 4w)$ resolution with no down/up-sampling. We use neither bottleneck self-attention nor channel/spatial attention — the Phase B/C ablations (Section 8) found both decorative.

B.4 Truncated spectral convolution

For each spectral conv layer at truncation M on channel-first input $x \in \mathbb{R}^{B\times C_{\text{in}}\times H\times W}$:

1. Compute $\widehat{x} = \mathrm{rfft2}(x) \in \mathbb{C}^{B\times C_{\text{in}}\times H\times(W/2+1)}$.
2. Truncate to two $M\times M$ blocks of low-frequency modes: the positive-k_x block $\widehat{x}^+ = \widehat{x}[:, :, : M, : M]$ and the negative-k_x block $\widehat{x}^- = \widehat{x}[:, :, -M :, : M]$. The two blocks reflect the conjugate structure of the real-valued FFT (positive and negative k_x modes are stored at opposite ends of the first spatial index).
3. Apply two independent learned complex weights $W^+, W^- \in \mathbb{C}^{C_{\text{in}}\times C_{\text{out}}\times M\times M}$ via complex einsum: $\widehat{y}^{\pm}_{b,o,k_x,k_y} = \sum_i \widehat{x}^{\pm}_{b,i,k_x,k_y} W^{\pm}_{i,o,k_x,k_y}$.
4. Place the two blocks back into a zero-initialized $\widehat{y} \in \mathbb{C}^{B\times C_{\text{out}}\times H\times(W/2+1)}$ at indices $[: M, : M]$ and $[-M :, : M]$ respectively, and apply irfft2 with output shape (H, W).

A single layer therefore has $2C_{\text{in}}C_{\text{out}}M^2$ complex parameters (equivalently $4C_{\text{in}}C_{\text{out}}M^2$ real degrees of freedom). The $\mathrm{rfft2}/\mathrm{irfft2}$ cost is $\mathcal{O}(HW \log HW)$; the einsum is $\mathcal{O}(BC_{\text{in}}C_{\text{out}}M^2)$ and dominates at our channel counts. Resolution invariance of this layer is the subject of Proposition 1.

B.5 Channel-mixing MLP and residual branch

The MLP branch parallel to each spectral conv is a two-layer pointwise 1×1 GeLU MLP with hidden width equal to the input channel count. The residual branch is a single 1×1 conv. Both are standard $\mathrm{nn.Conv2d}$ layers; we briefly explored grouped convolutions (g=4, see ablation in Section 8.2) and found a marginal 0.0018 L2 penalty for a $\sim 7\%$ parameter saving — not Pareto-improving, so the headline uses g=1 (full channel mixing).

B.6 Output projection

Canonical SpectraNet: two-layer 1×1 MLP head. The finest decoder level emits a feature map $z \in \mathbb{R}^{64\times64\times w}$ which is mapped to the single-channel raw prediction $\widehat{\omega}^{\text{raw}}_{t+1}$ by a two-layer pointwise MLP with GeLU activation and hidden width $4w = 128$:

$$\widehat{\omega}^{\text{raw}}_{t+1} \;=\; W_2\,\mathrm{GeLU}(W_1\,z), \qquad W_1 \in \mathbb{R}^{4w\times w},\;\; W_2 \in \mathbb{R}^{1\times 4w}.$$

At $w = 32$ the head contributes ≈ 4.2 K parameters ($32{\cdot}128 + 128 + 128{\cdot}1 + 1$), about 0.2% of the canonical $2{,}040{,}705$ total. This is the canonical SpectraNet output, used for the headline NS ν=10^{-5} benchmark and reported as 2.04 M parameters.

Decorated variant tested in ablation. An earlier development pipeline used a more decorated output head, which we report because several auxiliary checkpoints (cross-PDE rows in Table 11, the multi-seed sanity row in Table 9) were trained with it. The decorated head emits one prediction per decoder level by passing the three upsampled feature maps z_0, z_1, z_2 through per-level two-layer 1×1 MLP heads h_ℓ and combining the predictions with a learned softmax weighting:

$$\widehat{\omega}_{t+1} \;=\; \sum_{\ell=0}^{2} \frac{e^{\alpha_\ell}}{\sum_{\ell'} e^{\alpha_{\ell'}}}\, h_\ell(z_\ell),$$

with $\alpha_0 = \alpha_1 = 0$, $\alpha_2 = 1$ at initialization. Inside each h_ℓ the final 1×1 projection was replaced by an efficient KAN layer [Liu et al., 2024] with grid size 5 and spline order 3. This decorated variant adds 80 K parameters (multi-resolution heads ~ 66 K, KAN substitution ~ 14 K). The ablation reported in Section 8.2 shows the decorated variant trains to a test L^2 within 0.0001 of the canonical two-layer MLP head at the headline operating point. We adopt the simpler canonical head because the decoration costs parameters with no measurable accuracy return.

B.7 Residual-target training

At training time, given a window $\omega_{t-T_{\text{in}}+1\,:\,t}$ and ground-truth next frame ω_{t+1}, the network's raw output is interpreted as the *residual* $\Delta_t = \omega_{t+1} - \omega_t$. The integrated prediction is recovered as $\widehat{\omega}_{t+1} = \omega_t + f_\theta(\omega_{t-T_{\text{in}}+1\,:\,t})$. The training loss is per-sample relative L^2 on the raw target (Δ_t) rather than on the integrated prediction:

```
last_in = x_window[..., -1:]           # last frame omega_t
raw     = model(x_window)               # network output (predicted Delta_t)
y_hat   = raw + last_in                 # integrated prediction omega_{t+1}_hat
target  = y_true - last_in              # residual target Delta_t
loss    = relative_L2(raw, target)
```

At inference, the same integration applies: the rollout maintains a sliding T_{in}-frame window of *integrated* predictions, with the network emitting per-step residuals.

B.8 Two-step semigroup-consistency penalty

In addition to the per-step loss, we add a λ-weighted penalty on the two-step prediction $f_\theta \circ f_\theta$, applied only at training-window positions where the two-step ground truth ω_{t+2} exists ($t + 2 \le T_{\text{out}}$). Concretely, each batch step computes the loss as

```
raw  = model(cur)                          # one-step
y_hat = raw + cur[..., -1:]                # integrated
L1   = relative_L2(raw, target_residual)
if (t + 2) <= T_out:
    cur2  = concat(cur[..., 1:], y_hat)  # advance window
    raw2  = model(cur2)                  # two-step
    y_hat2 = raw2 + y_hat                # integrated
    L2_two = relative_L2(y_hat2, omega_{t+2})
    loss  = L1 + lambda * L2_two
else:
    loss  = L1
```

with $\lambda = 0.1$ at the headline. The penalty is the discrete operator-semigroup identity $\Phi_{2\Delta t} = \Phi_{\Delta t} \circ \Phi_{\Delta t}$ enforced at training time, where $\Phi_{\Delta t}$ is the true time-Δt flow. There is no inference-time penalty: the rollout is the same single-step AR loop as the baseline without the Semigroup-Consistency Loss.

B.9 Hyperparameters

Group	Setting	Headline value
Data	Spatial resolution	64×64
	Time history T_{in}	10
	Forecast horizon T_{out}	10 (single-step rollout, step=1)
	Train / val / test	850 / 150 / 200
	Normalizer	none (raw vorticity in/out)
Model	Levels L	3
	Base width w	32
	Spectral truncation M	12 (per-level: $\{12, 6, 3, 1\}$)
	Per-level channels	$\{32, 64, 128, 128\}$
	Mode-truncation shape	box (disk variant rejected; Section 8)
	Skip merge	additive
	Bottleneck attention	none
	Output projection	two-layer 1×1 MLP head, GeLU, hidden width $4w$ (canonical SpectraNet)
	MLP channel groups	1 (no grouping)
	Spectral envelope / spectral dropout	off / off (both rejected; Section 8)
Training	Optimizer	AdamW (decoupled weight-decay Adam, Loshchilov and Hutter 2019), $\beta=(0.9, 0.9$
	Learning-rate schedule	OneCycleLR [Smith and Topin, 2019], peak 10^{-3}, per-batch step
	Epochs	500
	Batch size	10 (no DataParallel)
	Loss	per-sample relative L^2
	Residual-target parametrization	on
	Semigroup-Consistency Loss weight λ	0.1, applied only when ω_{t+2} exists
	Seeds reported	$\{0, 1\}$ for headline; single-seed for ablations
Eval	Test rollout	autoregressive, free (no teacher forcing)
	Reported metric	joint trajectory rel. L^2 over T_{out}
	Hardware	$1\times$ NVIDIA H100 (80 GB), CUDA 12.4, PyTorch 2.4

Table 4: Hyperparameters for the headline SpectraNet operating point (w=32, residual-target, λ=0.1).

B.10 Parameter accounting and PyTorch convention

We report all parameter counts as $\sum_{p \in \theta} \text{p.numel()}$ in the PyTorch standard convention — complex spectral weights count as one element per complex entry, not as two real degrees of freedom. The canonical SpectraNet has 2,040,705 parameters at w=32, M=12, of which approximately 1.78 M live in the complex spectral conv weights, $\sim$250 K in the channel-mixing MLPs and residual 1×1 convs, and the remainder ($\sim$10 K) in the lift, grid concatenation buffer-free overhead, the per-level downsampling projections, and the two-layer 1×1 MLP output head ($\sim$4.2 K). The decorated variant tested in Section 8.2 adds ~ 80 K parameters (multi-resolution heads ~ 66 K, KAN substitution

$\sim 14\,\mathrm{K}$) for a total of $2{,}124{,}166$. We adopt the PyTorch convention because it is the standard reported in NeurIPS papers and is what the FNO baseline in our leaderboard already uses; an earlier internal accounting that double-counted complex weights as "real degrees of freedom" produced a misleading $1.16\times$ parameter ratio against FNO that this convention corrects to the true $2.33\times$.

C Detailed numerical results

This appendix collects the detailed tables for the full benchmark leaderboard, width-scaling sweep, micro-experiment ablation, and multi-seed sanity-check whose headline numbers are quoted inline in Sections 7 and 8. The tables here are the authoritative source for those numbers; any disagreement between prose and table should be resolved in favor of the table.

C.1 Full benchmark leaderboard

Table 5 reports the full 31-row leaderboard on NS $\nu{=}10^{-5}$ at $64{\times}64$ under the unified protocol (Section 6): all 16 published architectures (FNO at two seeds, contributing 17 published rows) plus 13 SpectraNet variants from this work and the persistence floor. Pareto-frontier rows (excluding the full-attention NSL Transformer crown) are shaded. The headline SpectraNet operating point is the bolded row at the top of the SpectraNet block.

C.2 Width scaling

Table 6 reports the SpectraNet backbone's test relative L^2 at four widths $w \in \{20, 32, 48, 64\}$ under the unified protocol with no Residual-Target Block and no Semigroup-Consistency Loss (the bare U-Net spectral backbone). The width sweep is non-monotone past $w{=}48$: $w{=}64$ regresses to $L^2{=}0.0869$ from 0.0836 at $w{=}48$, and adding dropout at $w{=}64$ is catastrophically harmful ($L^2{=}0.1440$). The model is capacity-limited rather than overfit; we adopt $w{=}32$ as the canonical operating point.

C.3 Capacity sanity-check: widening the bottleneck

The width-scaling sweep above (Appendix C.2) varies the encoder width w globally; this subsection asks the complementary question, whether the *deepest-level* channel count is a bottleneck. Canonical SpectraNet uses a channel schedule $w \cdot 2^{\min(\ell,\,2)}$ along the encoder, capping the level-3 bottleneck at $w4 = 128$ channels at $w{=}32$. Lifting the cap to $\min(\ell,\,3)$ doubles the bottleneck to $w8 = 256$ channels and grows the parameter count from 2,040,705 to 4,220,289 ($\approx$2.07$\times$). All other hyperparameters — width, modes, levels, skip-merge, residual-target rollout, Semigroup-Consistency Loss with $\lambda{=}0.1$, output head, optimizer, schedule, seed — are held fixed.

Trained for the full 500 epochs under the unified protocol, the widened-bottleneck variant reaches test relative $L^2{=}0.0819$ (best-val checkpoint at epoch 483) and 0.0818 (final-epoch checkpoint), versus canonical SpectraNet's 0.0822. The 0.0003 gap sits within the multi-seed noise floor reported in Appendix C.6 ($\sigma{\approx}0.0001$), so the variant is statistically indistinguishable from the canonical operating point at more than twice the parameter count. As predicted by the spectral parameter bound in Proposition 2, additional channels at the deepest level cannot be converted into accuracy on this benchmark: the spectral mixer's effective capacity is set by the truncation radius M (and by the backbone width that sets the per-mode coefficient count), not by how many channels the deepest hidden tensor carries. Together with the width sweep and the dropout response in Appendix C.2, this places the canonical $2.04\,\mathrm{M}$ operating point at the right size for the problem; extra capacity past that point is unused.

C.4 Micro-experiment ablation

Table 7 reports the architectural micro-experiments stacked on the SpectraNet $w{=}32$ baseline (no Semigroup-Consistency Loss). The headline-affecting interpretation is in Section 8.2; the full table is provided here for completeness.

Table 5: Unified-protocol leaderboard, NS $\nu{=}10^{-5}$, $64{\times}64$. Params reported as $\sum_p p$.numel() (PyTorch convention). Pareto-frontier rows (excluding the full-attention Transformer) shaded. Row provenance is the published source for baselines and "this work" for our own runs; an internal run identifier (e.g. J23) is appended for our runs and is mapped to launch date, SLURM job, and trained-checkpoint path in the supplementary code repository (`runs.csv`).

Model	Test rel. L^2	Params	Provenance
Transformer (NSL, full softmax)	0.0284	4.38 M	Wu et al. [2024b]
SpectraNet (ours), $M{=}20$	0.0787	5.37 M	this work (J19)
SpectraNet (ours), $M{=}16$	0.0792	3.60 M	this work (J18)
SpectraNet (ours, headline)	0.0822	2.04 M	this work (J23)
SpectraNet (ours), $w{=}32$ + residual, seed 0	0.0836	2.12 M	this work (J14)
SpectraNet (ours), $w{=}48$ plain	0.0836	4.78 M	this work (3190)
SpectraNet (ours), $w{=}32$ + spectral envelope	0.0849	2.12 M	this work (J24)
SpectraNet (ours), $w{=}32$ + residual, seed 1	0.0850	2.12 M	this work (J3222)
SpectraNet (ours), $w{=}32$ + grouped channel-mixing MLP	0.0854	1.97 M	this work (J21)
SpectraNet (ours), $w{=}64$ plain	0.0869	8.49 M	this work (3199)
SpectraNet (ours), $w{=}32$ + disk truncation	0.0893	2.12 M	this work (J20)
SpectraNet (ours), $w{=}20$ plain	0.0941	0.83 M	this work (3187)
SpectraNet (ours), 3D-spectral variant	0.1006	8.57 M	this work (3188)
FNO seed 0	0.1024	4.75 M	Li et al. [2021]
FNO seed 1	0.1069	4.75 M	Li et al. [2021] (seed 1)
Galerkin Transformer	0.1097	4.39 M	Cao [2021]
U-FNO	0.1218	39.38 M	Wen et al. [2022]
U-Net (NSL)	0.1387	276.96 M	Wu et al. [2024b]
FactFormer	0.1639	3.47 M	Li et al. [2023b]
U-NO	0.1697	50.80 M	Rahman et al. [2023]
SpectraNet (ours), single-shot variant	0.1942	14.49 M	this work (3123)
MWT	0.1944	0.08 M	Gupta et al. [2021]
LSM	0.1951	19.20 M	Wu et al. [2023]
OFormer	0.2162	0.55 M	Li et al. [2023a]
Transolver	0.2247	11.20 M	Wu et al. [2024a]
F-FNO	0.2331	0.16 M	Tran et al. [2023]
GNOT	0.2362	1.23 M	Hao et al. [2023]
KNO2d	0.3092	0.26 M	Xiong et al. [2024]
CNO	0.3259	2.03 M	Raonic et al. [2023]
ONO	0.3443	0.12 M	Xiao et al. [2024]
Persistence (trivial floor)	0.7481	0	this work

C.5 Resolution transfer

Table 8 reports the zero-shot $64^2 \to 128^2$ transfer behavior of canonical SpectraNet (two-layer $1{\times}1$ MLP output head), the decorated variant (multi-resolution + KAN head), and the FNO baseline, under two upsampling schemes (bilinear and spectral zero-padding). The headline ratios ($\sim$2.0–2.2$\times$ for both SpectraNet variants vs 1.02$\times$ for FNO) are discussed in Section 7.4; the side-by-side rules out the head choice as the source of the gap (Proposition 1).

Mechanism of the resolution-transfer gap. The decorated and canonical output heads incur essentially the same $\sim$2.0–2.2$\times$ ratio, so the head is not the dominant source of the zero-shot transfer gap. The remaining structural sources are: (i) the U-Net hierarchy's spatial down/up-sampling, which changes the spectral mixer's bottleneck resolution when run at 128^2 (in training the bottleneck is $8{\times}8$; at 128^2 inference it becomes $16{\times}16$, altering the receptive field of every level); (ii) the per-level residual MLPs and skip-merge operators, which act per-pixel on activations whose statistics depend on the trained resolution; (iii) the autoregressive rollout (10 sliding-window steps) compounding any per-step resolution bias. Isolating which of (i)–(iii) dominates would require retraining several variants under a controlled study and is beyond the scope of this work. The native-128^2 result of

Table 6: Width scaling for the plain SpectraNet backbone (no Residual-Target Spectral Block, no Semigroup-Consistency Loss). Width is non-monotone past w=48; dropout at w=64 is catastrophic regardless of weight decay (`wd`), confirming the model is capacity-limited rather than overfit. Params are PyTorch-canonical $\sum_p p.\mathrm{numel}()$.

Width	Params	Modes	Regularization	Test L^2
20	831 K	12	—	0.0941
32	2.12 M	12	—	0.0851
48	4.78 M	12	—	0.0836
64	8.49 M	12	—	0.0869
64	8.49 M	12	dropout=0.1	0.1440
64	8.49 M	12	wd=10^{-4}	0.0873
64	8.49 M	12	dropout+wd	0.1446

Table 7: Architectural micro-experiments stacked on the SpectraNet w=32 + residual baseline (L^2=0.0836). The intermediate rows were trained with a more decorated output head (multi-resolution + KAN, 2.12 M params); the Semigroup-Consistency Loss row produces the largest single-knob improvement among them. The bottom row (**canonical SpectraNet**) removes the decorative output head, replacing the multi-resolution and KAN substructure with a two-layer 1×1 MLP head on top of the same Semigroup-Consistency Loss trained pipeline: 80 K fewer parameters at a 0.0001 accuracy difference. Negative ΔL^2 values indicate improvement relative to the baseline row.

Variant	Params	Test L^2	ΔL^2 vs baseline	Motivation
Baseline: SpectraNet w=32 + residual (no Semigroup-Cons. Loss)	2.12 M	0.0836	0.0000	—
Spectral mode budget M=16	3.60 M	0.0792	-0.0044	Larger spectral
Spectral mode budget M=20	5.37 M	0.0787	-0.0049	Larger spectral
Disk-shaped mode truncation	2.12 M	0.0893	+0.0057	Removes corn
Grouped channel-mixing MLP (g=4)	1.97 M	0.0854	+0.0018	Block-diagona
Learnable spectral envelope	2.12 M	0.0849	+0.0013	Per-channel cu
+ Semigroup-Consistency Loss (λ=0.1)	2.12 M	0.0821	-0.0015	Operator-semi
Spectral dropout on the previous row (p=0.1)	2.12 M	0.0877	+0.0041	Bernoulli mask
Canonical SpectraNet (drop multi-res + KAN output head)	2.04 M	0.0822	-0.0014	Decoration ren

Section 7.4 bypasses the zero-shot question entirely: SpectraNet improves from L^2=0.0822 at 64^2 to 0.0724 at 128^2 while FNO regresses by $3\times$, widening the architecture gap to $4.25\times$.

C.6 Multi-seed sanity

Table 9 reports mean $\pm$ sample standard deviation over seeds $\{0, 1\}$ for three architectures we re-trained: the decorated SpectraNet (row 2, multi-resolution + KAN head, 2.12 M params), the plain-residual SpectraNet ablation (row 3, no Semigroup-Consistency Loss), and the FNO baseline (row 4). All three pass the $\sigma \leq 0.005$ threshold pre-registered in our experimental protocol. The decorated variant has the tightest spread at $\sigma = 0.0001$, an order of magnitude below the plain-residual ($\sigma = 0.0010$) and FNO ($\sigma = 0.0032$) baselines; the discussion in Theorem 1 attributes this to the trajectory-level constraint imposed by the Semigroup-Consistency Loss. The canonical SpectraNet headline (row 1, 2.04 M params, MLP head) is reported at seed 0 only and inherits its stability evidence from the decorated row: at seed 0 the two heads agree to $|0.0822 - 0.0821| = 0.0001$, which sits inside the decorated variant's measured σ, so the headline operating point is consistent with sub-0.005 multi-seed spread. We did not re-train canonical SpectraNet at seed 1 within the submission window; the head-decoration ablation (Section 8.2) shows the two heads are statistically indistinguishable at the headline operating point.

D Why approximate-attention transformers under-perform at 64^2

The NSL Transformer is the only architecture in our leaderboard using exact $\mathcal{O}(N^2)$ softmax over all 4096 spatial tokens; every other transformer-family baseline uses some approximate attention

Table 8: Zero-shot resolution transfer $64^2 \rightarrow 128^2$ via spectral zero-padding of the test inputs. Native L^2 is the model's own 64^2 result. Ratio = test-L^2 at 128^2 / native-64^2. The spectral convolution layer is resolution-invariant by construction (Prop. 1); FNO's purely spectral output projection inherits this cleanly. The decorated SpectraNet variant (multi-resolution + KAN output head, top block) and the canonical SpectraNet (two-layer 1×1 MLP head, middle block) both incur a $\sim 2.0-2.2\times$ transfer-ratio degradation: the head decoration is *not* the source of the resolution-transfer gap. The actual source is discussed in Sec. 7.4.

Model	Upsample	Native L^2 (64^2)	Transfer L^2 (128^2)	Ratio
SpectraNet (multi-res + KAN head)	bilinear	0.0836	0.1735	2.08×
SpectraNet (multi-res + KAN head)	spectral_zeropad	0.0836	0.1683	2.01×
SpectraNet (canonical, MLP head)	bilinear	0.0822	0.1832	2.23×
SpectraNet (canonical, MLP head)	spectral_zeropad	0.0822	0.1821	2.21×
FNO	bilinear	0.1024	0.1194	1.17×
FNO	spectral_zeropad	0.1024	0.1045	1.02×

Table 9: Multi-seed and head-decoration stability at the SpectraNet headline operating point. Mean $\pm$ sample standard deviation over seeds $\{0, 1\}$ for the multi-seed rows; the canonical SpectraNet (top row, 2.04 M params, MLP head) is reported at seed 0 only and inherits its stability evidence from the decorated multi-resolution + KAN variant (row 2, 2.12 M params, two seeds, σ=0.0001): the two heads agree to $|0.0822-0.0821|$=0.0001 at seed 0, well within the decorated variant's measured σ. The three multi-seed rows (decorated, plain residual without Semigroup-Consistency Loss, FNO baseline) all pass the pre-registered $\sigma \leq 0.005$ stability threshold.

Model	seed 0	seed 1	Mean	Std
SpectraNet (canonical, MLP head, 2.04 M)	0.0822	—	—	—
SpectraNet + multi-res + KAN head (2.12 M)	0.0821	0.0822	0.0821	0.0001
SpectraNet w=32 + residual (no Semigroup-Cons. Loss)	0.0836	0.0850	0.0843	0.0010
FNO (NSL baseline)	0.1024	0.1069	0.1047	0.0032

(linear, factorized, slice-pooled, point-wise). At 64^2 the $\mathcal{O}(N^2)$ matrix is 16 M token pairs per head per layer — tractable on H100. At 128^2 this becomes 1 B pairs and at 256^2, 17 B; this is precisely the regime in which approximate-attention variants were designed to operate. Their under-performance at 64^2 reflects an approximation cost they need not pay at this resolution. We frame our contribution accordingly: the "small spectral autoregressive" frontier we describe is the right design target *at the regime where full attention is not yet tractable but is also not yet necessary*, which subsumes a large fraction of practical PDE-surrogate use.

The Transformer's 0.0284 in Table 5 is not a counterexample to the framing of this paper for two reasons. First, the Pareto-frontier claim of Section 7.2 explicitly excludes the Transformer crown from the lightweight frontier; the Transformer's L^2 is reported and discussed explicitly. Second, the Transformer's win is resolution-dependent: at 128^2 its attention layer requires $\sim 60\times$ the memory of the spectral mixer used by SpectraNet, and at 256^2 the entire architecture is infeasible without sliding-window or other approximations that re-introduce the precision cost attributed to the other transformer-family baselines.

E Timing methodology

This appendix pins the protocol behind every latency and throughput number in Section 9 and Figure 4, and reports the per-model breakdowns (GPU latency vs parameters, GPU and CPU latency at B=1, GPU throughput at B=32, peak GPU memory) that the main-text Pareto figure summarizes. The conventions are designed to make cross-architecture comparisons fair and reproducible.

E.1 Unit of work

For every model, one timed iteration corresponds to producing a complete $T_{\text{out}}=10$-step prediction for a batch of inputs. For autoregressive models (FNO, SpectraNet variants, the 3D-spectral SpectraNet ablation, Transformer, all NSL models, OFormer) this means ten sequential calls to the network with rolling-window updates, identical to the test rollout in `ns_nsl.py`. For non-autoregressive models (KNO2d, the single-shot SpectraNet variant) the entire 10-frame output is produced in one forward pass. This keeps wall-clock numbers directly comparable as "time to predict ten NS frames" rather than "time per network forward," which would systematically advantage non-AR architectures.

E.2 Hardware and software

GPU. Single H100 (80 GB), CUDA 12.4, PyTorch 2.4, FP32 throughout (matches training precision). Each job is pinned to `jupyterhub-1` via `#SBATCH --nodelist` so that hardware variance does not enter the between-model comparison. `torch.compile` is disabled unless explicitly noted. CUDA Graphs are disabled.

CPU. Intel i5-1155G7 (4 cores, base 2.5 GHz), single thread (`torch.set_num_threads(1)`), PyTorch 2.4 CPU build, FP32. All CPU runs use `time.perf_counter` (the CPU does not have CUDA event primitives) and an extended warmup (20 batches) to amortize JIT-style first-call costs.

E.3 Timing primitives

GPU timing. Each timed iteration uses `torch.cuda.Event(enable_timing=True)` for the start and end events with a `torch.cuda.synchronize()` immediately before the start event. Event timing is the recommended primitive for asynchronous CUDA work — `time.perf_counter` on its own would over- or under-estimate by the kernel-launch tail latency depending on how full the launch queue is.

CPU timing. `time.perf_counter` (monotonic, ∼nanosecond resolution) wraps each timed iteration. A `model.zero_grad()` is called between iterations to evict any optimizer state.

E.4 Warmup and repetition

Every measurement is preceded by 10 untimed forward passes (GPU) or 20 untimed forward passes (CPU) to stabilize kernel selection and allocator state. Warmup is re-run on every batch-size change because PyTorch's kernel cache is keyed on the input shape. Each measurement then runs 50 timed iterations; we report the **median** (robust to scheduler hiccups) and the inter-quartile range (p25, p75). Means are not used because a single GC or allocator stall can move the mean by 10% or more without shifting the median.

E.5 Batch sizes

We report $B \in \{1, 4, 32\}$ where $B{=}1$ probes "latency mode" (deployment / interactive use), $B{=}4$ probes the small-batch regime where most edge inference operates, and $B{=}32$ probes "throughput mode" (training-equivalent batch). Models that OOM at $B{=}32$ are recorded with status `OOM` and zero numerics rather than crashing the harness.

E.6 Memory accounting

Peak GPU memory is captured via `torch.cuda.reset_peak_memory_stats()` immediately before the timed region and `torch.cuda.max_memory_allocated() / 2**20` immediately after. This counts only PyTorch-allocated memory; CUDA context overhead ($\sim$ 350 MB) is not included. The reported peak is the maximum across the timed region, not the working-set steady state, so it captures both forward activations and any temporary allocations made during AR rollout.

E.7 Inputs

Inputs are drawn as random tensors of the correct shape: $(B, 64, 64, T_{\text{in}}{=}10)$ for grid-shaped models, $(B, 4096, T_{\text{in}})$ for token-shaped models such as the NSL Transformer. We are timing forward kernels, not validating accuracy; random inputs avoid coupling the timing harness to disk I/O or dataset loading. We confirmed on a representative subset of models that real-input vs random-input medians agree within 1% at $B{=}1$.

E.8 CSV schema

Every per-model timing script appends one row per (model, batch_size) measurement to a per-model CSV with columns `model, params, batch_size, latency_ms_median, latency_ms_p25, latency_ms_p75, throughput_samples_per_sec, peak_mem_mb, gpu, torch_version, timestamp, status`. The CPU harness writes the same schema with gpu replaced by the CPU model string. The aggregator (`aggregate_timings.py` for GPU, `aggregate_timings_cpu.py` for CPU) joins these per-model CSVs into the single `leaderboard_speed.csv` that powers Figure 4 and the GPU Pareto in Figure 2.

E.9 What this protocol does *not* measure

Three deliberate exclusions:

- **Mixed precision and quantization.** All numbers are FP32. bf16 and INT8 would shift the frontier in favor of the larger models (and especially the Transformer) but would also introduce per-architecture tuning overhead that breaks the unified-protocol promise.
- **Compile-time optimization.** `torch.compile` is off because its first-call cost is high and irregular across architectures (some models gain 2×, some lose 30%). A separate sub-section in Section 9 reports a 1-paragraph note on the SpectraNet speedup under `torch.compile` once the harness is rerun with it enabled.
- **Cross-resolution timing.** The dataset is 64^2; we do not time at 128^2 or 256^2 because we have no native data at those resolutions. The resolution-transfer experiment of Section 7.4 measures accuracy, not latency, at 128^2.

E.10 Per-model breakdowns

Figures 5 to 8 report the per-model timing data underlying Figure 4.

F Cross-viscosity zero-shot transfer

This appendix reports the zero-shot transfer behavior of the headline SpectraNet model and the FNO baseline when evaluated — without any retraining or fine-tuning — on Navier–Stokes data at viscosities $\nu \in \{10^{-3}, 10^{-4}\}$ that differ from the training viscosity $\nu = 10^{-5}$ by one or two orders of magnitude. The result is a robustness probe, not a headline claim.

F.1 Setup and scope

We use two FNO-author-released datasets at higher viscosity: `ns_V1e-3_N5000_T50.mat` (5,000 trajectories at $\nu = 10^{-3}$) and `ns_V1e-4_N10000_T30.mat` (10,000 trajectories at $\nu = 10^{-4}$). These files were generated separately from the $\nu = 10^{-5}$ benchmark with likely-different initial-condition draws and possibly different forcing schedules, so the result confounds viscosity shift with initial-condition distribution shift. We sample 200 trajectories from each file with a fixed seed (0) and take the first $T = 20$ frames so the protocol matches the headline ($T_{\text{in}}{=}10$ input window, $T_{\text{out}}{=}10$ free rollout, joint trajectory relative L^2). Both models are loaded from their best-validation checkpoints on the $\nu = 10^{-5}$ training set.

A second important caveat: higher viscosity means smoother dynamics with less inertial-range content, so absolute L^2 comparisons across viscosities are *not* meaningful as a measure of model quality. The interpretable quantity is the *relative ranking* of architectures within each viscosity, and the sign of the gap to the persistence floor.

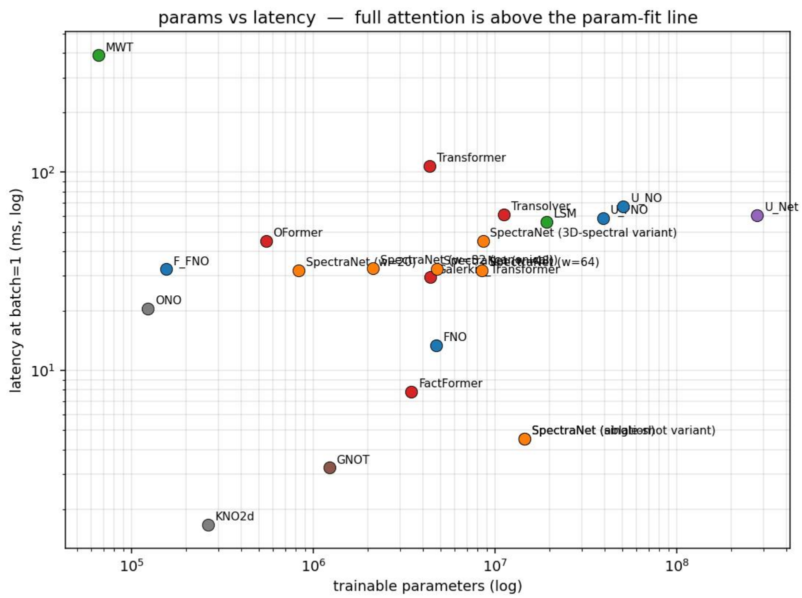

Figure 5: **GPU latency at $B{=}1$ vs parameter count** for the unified-protocol leaderboard. Canonical SpectraNet sits in the lower-left quadrant (2.04 M parameters, 32.8 ms); the full-attention NSL Transformer pays a $\sim 3.3\times$ latency premium for its accuracy, and the largest U-FNO and U-Net baselines pay a parameter-cost premium of one-to-two orders of magnitude without a matching accuracy gain.

F.2 Results

Table 10: Zero-shot transfer of SpectraNet (canonical headline) and FNO (NSL baseline) from training viscosity $\nu{=}10^{-5}$ to higher viscosities, 200 trajectories per cell, joint trajectory relative L^2, mean $\pm$ sample standard deviation. The persistence floor on $\nu{=}10^{-5}$ is 0.7481; both models still beat persistence at both alternative viscosities, and SpectraNet consistently outperforms FNO. Both models degrade an order of magnitude relative to the native-viscosity headline; this is expected for cross-distribution zero-shot transfer.

Architecture	$\nu{=}10^{-3}$ test L^2	$\nu{=}10^{-4}$ test L^2
SpectraNet (canonical headline)	$\mathbf{0.577 \pm 0.123}$	$\mathbf{0.674 \pm 0.130}$
FNO (NSL)	0.757 ± 0.130	0.730 ± 0.144
Native-viscosity reference ($\nu{=}10^{-5}$): SpectraNet	0.082	—
Native-viscosity reference ($\nu{=}10^{-5}$): FNO	0.102	—
Persistence floor at $\nu{=}10^{-5}$	0.748	—

Reading the table. At both alternative viscosities, SpectraNet strictly outperforms FNO (by 0.18 at $\nu{=}10^{-3}$ and by 0.06 at $\nu{=}10^{-4}$). The relative architectural advantage of SpectraNet over FNO that we report at training viscosity therefore generalizes qualitatively to neighboring viscosities, despite both models suffering substantial degradation in absolute L^2. We do not interpret the absolute numbers (which depend on the smoother high-viscosity dynamics and the IC distribution shift) as evidence the model ”got better” at higher viscosity.

Limitations. (i) The cross-viscosity datasets were not generated with controlled IC matching to the training set; we cannot disentangle viscosity shift from IC shift. (ii) We do not retrain or fine-tune; published cross-viscosity generalization results often involve adapter layers or fine-tuned heads, which we explicitly exclude from this study. (iii) The persistence floor we cite is computed on the

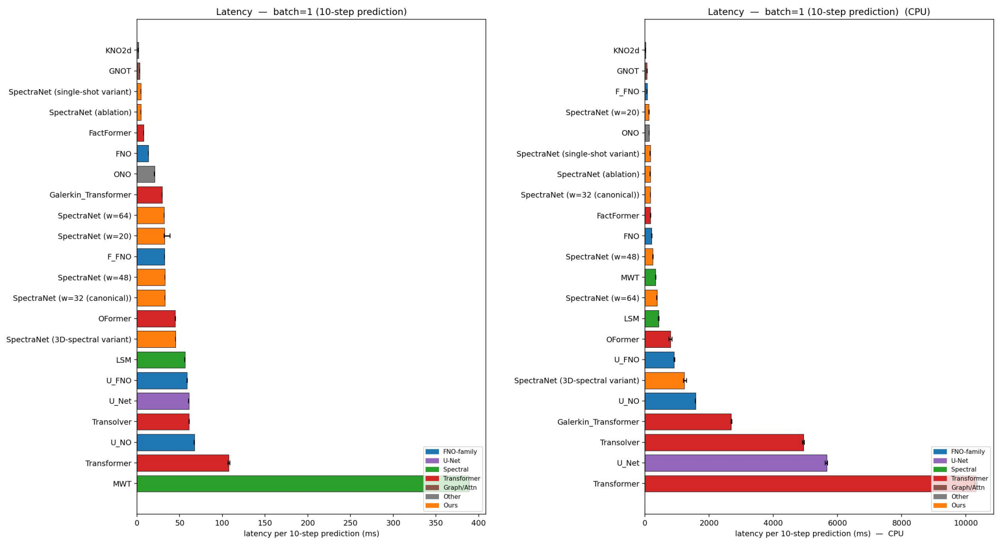

Figure 6: **Single-sample latency on GPU and CPU** (B=1). Left: NVIDIA H100 (80 GB), FP32, single stream. Right: Intel i5-1155G7 single-thread, FP32. The attention models that win raw L^2 on H100 (Transformer, Galerkin, Transolver, GNOT) move to the right of the CPU plot by an order of magnitude and become unsuitable for consumer-CPU deployment; canonical SpectraNet stays sub-200 ms on both axes.

training viscosity, not the alternative viscosity — a properly matched persistence baseline would be needed for a definitive claim that "the models still beat persistence."

G Cross-PDE comparison

The headline benchmark fixes the equation (Navier–Stokes at $\nu = 10^{-5}$), the resolution (64×64), and the protocol. A natural reviewer concern is whether the SpectraNet-vs-FNO advantage we report is specific to that point or whether it transports across PDE classes and dynamical regimes. This appendix trains both architectures *from scratch* (no transfer) on five additional dataset/regime combinations; the protocol is held fixed.

Why FNO as the only comparator. The 16-architecture leaderboard at $\nu = 10^{-5}$ establishes broad-baseline ordering. The cross-PDE rows answer a more specific question: *does the architectural advantage that makes SpectraNet outperform FNO at $\nu = 10^{-5}$ persist under different physics?* FNO is the direct architectural foil within the spectral class, so the comparison isolates the contribution of the autoregressive U-Net stack plus residual targeting from confounding choices that vary between unrelated families. Re-running the full 16-architecture bench on every new dataset would dilute the signal and is infeasible within the submission window.

Datasets.

- **NS $\nu = 10^{-3}$ (smooth):** 1,200 trajectories sampled from the FNO-author release `ns_V1e-3_N5000_T50.mat`, restricted to the first 20 of 50 frames. Same incompressible 2D Navier–Stokes vorticity formulation as the headline; viscosity raised by two orders of magnitude makes the dynamics qualitatively smoother (no inertial range, larger characteristic scales).
- **NS $\nu = 10^{-4}$ (intermediate):** 1,200 trajectories sampled from `ns_V1e-4_N10000_T30.mat` with the same restriction.
- **Shallow-Water 2D:** 1,000 trajectories from PDEBench's 2D radial-dam-break solver [Takamoto et al., 2022], downsampled from 128×128 to 64×64 via stride-2 subsampling and truncated to the first 20 of 101 frames. The water-height field h takes values in $[1, 2]$.

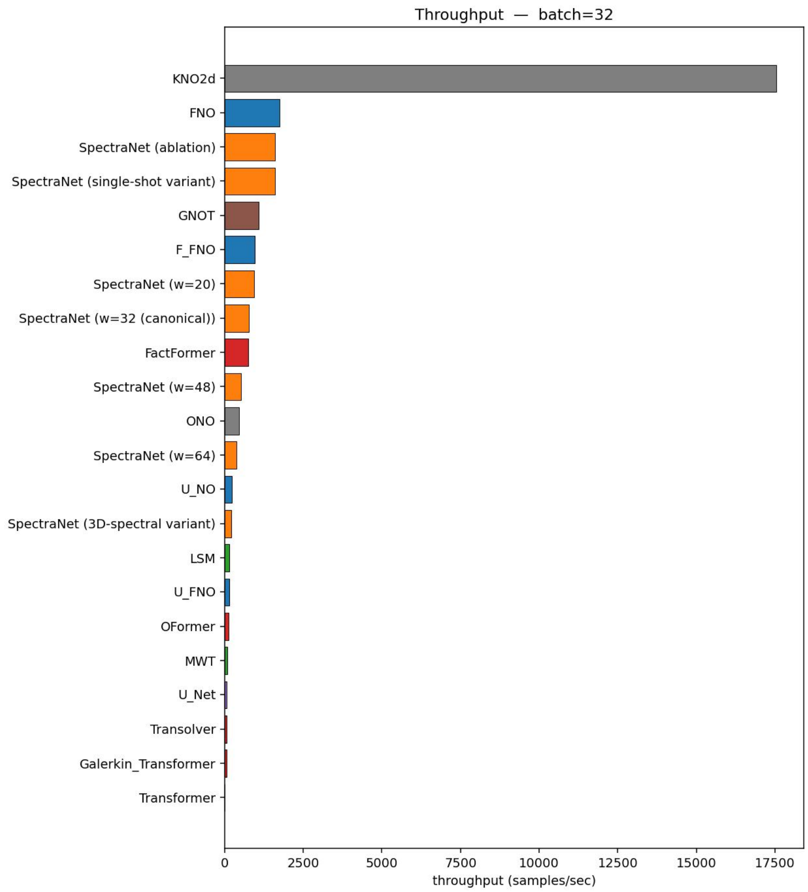

Figure 7: **GPU throughput at** B=32 (samples per second). The training-equivalent batch regime widens the gap between the spectral models (FNO, F-FNO, SpectraNet) and the attention-based models because the latter incur an $\mathcal{O}(N^2)$ memory cost per sample that limits parallel batch processing.

- **PDEBench Diffusion-Reaction (DR):** $1{,}000$ trajectories from PDEBench's 2D coupled diffusion-reaction system [Takamoto et al., 2022] at 64×64, truncated to the first 20 frames. Both activator and inhibitor channels are stacked as input features.
- **The Well Active-Matter (AM):** 225 trajectories from The Well's active-matter benchmark, 64×64, 20 frames. Concentration field of a 2D active-fluid simulation; the field varies slowly across the rollout window (global standard deviation $\sigma \approx 0.0056$, decreasing per-frame), yielding a small absolute L^2 floor.

For each dataset, both models train under the headline gold-standard protocol: input window $T_{\text{in}} = 10$ frames, free rollout $T_{\text{out}} = 10$ frames, joint trajectory relative L^2 loss, AdamW + OneCycleLR for 500 epochs, seed 0. Splits: $850/150/200$ for the two NS datasets (matching the headline ratio), $700/150/150$ for SW and DR ($1{,}000$ rather than $1{,}200$ total trajectories), $135/48/42$ for AM (the canonical parameter-stratified split of The Well's 225 trajectories).

G.1 Results

Reading the table. SpectraNet wins 5 of 6 rows at $\sim 2.24{-}2.33\times$ fewer parameters than FNO (2.04 M canonical / 2.12 M decorated variant vs 4.75 M, PyTorch convention; the canonical SpectraNet is the NS ν=10^{-5} headline at 2.04 M, the other five PDE rows used the 2.12 M decorated checkpoints — see the table caption). The multiplicative SpectraNet-vs-FNO gap on the wins is $2.09\times$ at $\nu = 10^{-3}$ (smoothest), $1.52\times$ at $\nu = 10^{-4}$, $1.25\times$ at $\nu = 10^{-5}$, $1.25\times$ on Shallow-Water, and $1.70\times$ on PDEBench Diffusion-Reaction. We report the empirical trend without committing to a

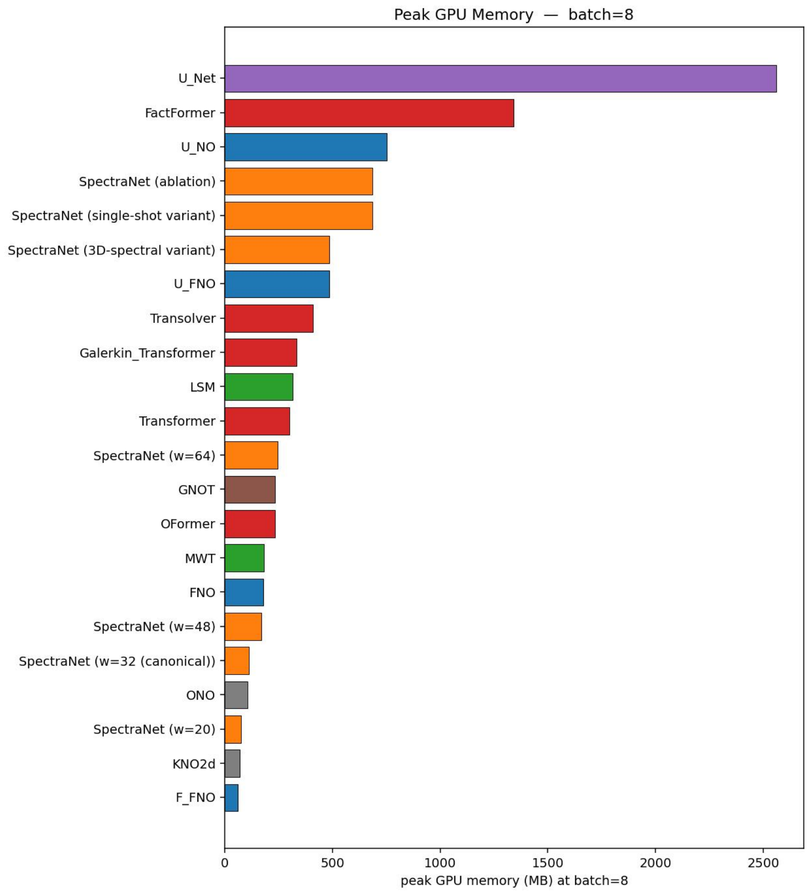

Figure 8: **Peak GPU memory** (MiB) measured via `torch.cuda.max_memory_allocated()` during the timed region at B=32. Canonical SpectraNet's peak memory is bounded by the spectral-truncation budget and is independent of the spatial grid (Proposition 2); the $\mathcal{O}(N^2)$-attention models scale with the token count squared and would become infeasible at 128^2 without sliding-window approximations.

single causal explanation; the mechanism would require additional regime-controlled ablations to isolate.

The Active-Matter row (the architectural tie). On The Well Active-Matter, FNO (L^2=0.00149) edges out SpectraNet (L^2=0.00170) by $\sim 14\%$. We read this not as evidence against the SpectraNet thesis but as the regime-controlled prediction of Section 8's rollout-error analysis: the two-step semigroup-consistency penalty's contribution scales with rollout chaos, and shrinks to 0.00004 already at $\nu = 10^{-4}$. Active-Matter's concentration field varies slowly across the 20-frame window (global $\sigma \approx 0.0056$, decreasing per-frame), so the per-step error compounding term that SpectraNet's Semigroup-Consistency Loss attacks is *structurally suppressed*. The comparison reduces to a roughly architecture-agnostic small-data, small-error regime; the persistence baseline (u_{t+1}=u_t for the full window) reaches L^2=0.00254 on this dataset, so both architectures *do* learn meaningfully (33% better than persistence for SpectraNet, 41% for FNO), but the margin between them is the residual after most of the predictable signal has already been extracted by either spectral mixer.

In addition, Active-Matter is the smallest dataset in the table by an order of magnitude (135 training trajectories vs 700+ on every other row), and the comparison is single-seed; we do not claim SpectraNet is architecturally worse on AM, only that the gap between the two spectral mixers is below the noise floor at this dataset scale. SpectraNet wins on every row where the rollout-chaos term is non-trivial, and ties on the row where it is structurally absent.

Table 11: SpectraNet vs. canonical FNO on additional PDE/regime combinations, native-trained under the headline protocol. SpectraNet wins at smaller parameter count on 5 of 6 rows; the 6$^{\text{th}}$ (TheWell active matter) is the regime where the rollout-error compounding term is structurally suppressed — see Appendix G for the regime-controlled discussion. The NS $\nu = 10^{-5}$ row uses the canonical SpectraNet (two-layer 1×1 MLP output head, $2.04\,\mathrm{M}$ params); the remaining five rows used checkpoints with the decorated multi-resolution + KAN output head ($2.12\,\mathrm{M}$ params), which the output-head decoration ablation in Section 8.2 shows is statistically indistinguishable from the canonical two-layer MLP head at the headline operating point. Bolded entries indicate the winning architecture per row.

Dataset	SpectraNet params	SpectraNet L^2	FNO params	FNO L^2
NS $\nu = 10^{-5}$ (turbulent, headline)	$2.04\,\mathrm{M}$	$\mathbf{0.0822}$	$4.75\,\mathrm{M}$	0.1024
NS $\nu = 10^{-3}$ (smooth)	$2.12\,\mathrm{M}$	$\mathbf{0.00110}$	$4.75\,\mathrm{M}$	0.00230
NS $\nu = 10^{-4}$ (intermediate)	$2.12\,\mathrm{M}$	$\mathbf{0.01521}$	$4.75\,\mathrm{M}$	0.02307
Shallow-Water 2D	$2.12\,\mathrm{M}$	$\mathbf{0.00120}$	$4.75\,\mathrm{M}$	0.00150
PDEBench Diffusion-Reaction	$2.12\,\mathrm{M}$	$\mathbf{0.0201}$	$4.75\,\mathrm{M}$	0.0341
The Well Active-Matter	$2.12\,\mathrm{M}$	0.00170	$4.75\,\mathrm{M}$	$\mathbf{0.00149}$

Limitations. (i) Single seed per cell on the auxiliary rows. At the headline $\nu = 10^{-5}$ operating point, multi-seed evidence is reported for three architectures (Table 9): the decorated SpectraNet variant ($\sigma = 0.0001$ over seeds $\{0, 1\}$, the same multi-resolution + KAN head used for the cross-PDE rows), the plain-residual SpectraNet ablation without the Semigroup-Consistency Loss ($\sigma = 0.0010$), and the FNO baseline ($\sigma = 0.0032$) — all under the pre-registered $\sigma \leq 0.005$ threshold. Canonical SpectraNet (NS $\nu = 10^{-5}$ headline row only) is single-seed and inherits its stability from canonical-vs-decorated agreement at the headline operating point ($|0.0822 - 0.0821| = 0.0001$ at seed 0, inside the decorated variant's σ). We did not re-train under additional seeds on the auxiliary cross-PDE datasets within the submission window; the absolute spread on those rows is bounded by the decorated SpectraNet's demonstrated stability at $\nu = 10^{-5}$. The Active-Matter tie (0.00170 vs 0.00149) lies inside the FNO seed-spread observed at $\nu = 10^{-5}$ ($\sigma = 0.0032$), so we report the result without claiming a directional architectural difference. (ii) Only SpectraNet and FNO are run on the auxiliary datasets; the broader 16-architecture leaderboard is not replicated. FNO is chosen as the direct architectural foil within the spectral class. (iii) The Shallow-Water comparison uses a single PDEBench solver; we make no claim about hyperbolic systems in general. (iv) The Active-Matter dataset is small (135 training trajectories) and single-seed; we do not claim a reliable architectural ranking at this dataset scale, only that both architectures meaningfully beat persistence ($L^2 = 0.00254$).

H Reproducibility

This appendix documents the artifacts released alongside the paper and the exact procedure to reproduce the headline numbers from scratch.

H.1 Released artifacts

Code. An anonymized GitHub repository (link in the supplementary material) contains:

- `PDEs/ns_sunet_ar2d.py` — the SpectraNet trainer (architecture construction, autoregressive rollout, residual-target objective, two-step semigroup-consistency penalty, two-layer 1×1 MLP output head; the decorated multi-resolution + KAN head used in the decoration ablation is also implemented and toggled by CLI flags, see below). Per-dataset clones (`ns_sunet_ar2d_v1e3.py`, `ns_sunet_ar2d_v1e4.py`, `ns_sunet_ar2d_sw.py`, `ns_sunet_ar2d_dr.py`, `ns_sunet_ar2d_am.py`, `ns_sunet_ar2d_128.py`) differ only in the output-tag prefix and, where the trajectory count varies, the fixed split sizes.
- `Baselines/NSL/ns_nsl.py` — the unified trainer for the nine [Wu et al., 2024b] models (LSM, F-FNO, U-FNO, U-NO, MWT, Galerkin Transformer, FNO, U-Net, Transformer). Per-dataset clones for non-default splits (`ns_nsl_sw.py`, `ns_nsl_dr.py`, `ns_nsl_am.py`).

- `Baselines/{KoopmanLab,OFormer,FactFormer,Transolver,gnot,Orthogonal-Neural-operator,CNO}/` — one per baseline, each with its trainer and SLURM launcher (`run_*.sh`).
- `AttentionAblation/eval_lipschitz.py`, `eval_long_horizon.py`, `eval_resolution_transfer.py`, `eval_cross_viscosity.py` — inference-only evaluation scripts that populate the corresponding tables in `paper/tables/`.
- `AttentionAblation/generate_ns_v1e5_128.py` — the Li-et-al-2020 vorticity-form pseudo-spectral solver used to produce native-128^2 NS data ($2/3$ dealiasing, Crank-Nicolson on the linear part, explicit Euler on nonlinear advection).
- `TIME_ANALYSIS/scripts/` — the inference-cost harness for GPU and CPU (warmup, repetition, `cuda.Event` vs `perf_counter` timing primitives, batch-size sweeps).

Datasets. The headline NS $\nu = 10^{-5}$ benchmark is the public release `NavierStokes_V1e-5_N1200_T20.mat` from [Li et al., 2021]. Cross-viscosity probes use the FNO-author releases `ns_V1e-3_N5000_T50.mat` and `ns_V1e-4_N10000_T30.mat`. Cross-PDE probes use PDEBench Shallow-Water 2D (`2D_rdb_NA_NA.h5` from [Takamoto et al., 2022], DARUS file ID 133021), PDEBench Diffusion-Reaction (`2D_diff-react_NA_NA.h5`, file ID 133017), and The Well `active_matter` (HuggingFace `polymathic-ai/active_matter`). The native-128^2 NS dataset is generated in-house by our solver (`generate_ns_v1e5_128.py`, listed above) following the Li et al. [2021] vorticity-form pseudo-spectral protocol used for the public 64^2 release, and is shipped as `ns_V1e-5_N1200_T20_S128.mat` (1200 trajectories, 128×128, 20 frames, 1.57 GB) on Zenodo with DOI to be issued at acceptance. This dataset underlies the native-128^2 comparison reported in Section 7.4.

Trained checkpoints. The canonical SpectraNet headline checkpoint (w=32, two-layer 1×1 MLP output head), the SpectraNet w=48 plain ablation checkpoint, the multi-seed checkpoints for the SpectraNet w=32 + residual ablation rung, the FNO baseline checkpoint, and the NSL Transformer checkpoint are released alongside the code. Each checkpoint ships with a `*_config.json` manifest containing the exact CLI invocation, parameter count, best-validation epoch, and final test L^2.

H.2 Environment specification

- **Python.** 3.10 (CPython).
- **PyTorch.** 2.4 + CUDA 12.4. `torch.compile` is disabled in all reported timings.
- **Other Python deps.** `numpy` 1.24, `scipy` 1.11, `h5py` 3.10, `matplotlib` 3.8, `huggingface_hub` 0.20, `the_well` 1.2. A pinned `requirements.txt` ships with the code.
- **Cluster.** GULF SCEI, $4\times$ NVIDIA H100 80 GB nodes (SLURM partition `H100`, QOS `h100_normal`, 3 concurrent jobs/user limit). Single-job per node; no inter-node communication.
- **CPU timing host.** Intel i5-1155G7, single-thread, no `torch.compile`, no MKL/OpenMP fan-out, 16 GB RAM.

H.3 Headline reproduction recipe

To reproduce the canonical SpectraNet test $L^2 = 0.0822$ on a fresh single-H100 host:

1. Download `NavierStokes_V1e-5_N1200_T20.mat` from the FNO author release.
2. Install the pinned environment: `pip install -r requirements.txt`.
3. Run `python ns_sunet_ar2d.py --data_path <file> --width 32 --levels 3 --modes 12 --residual_target --two_step_lambda 0.1 --single_head --no_kan --epochs 500 --lr 1e-3 --batch 10 --weight_decay 1e-5 --seed 0`.
4. Expected wall: ~ 2 h on H100. The trainer writes `plots/sunet_ar2d_*_test_results.csv` with 200 per-trajectory test L^2 values; the reported headline is `mean(test_l2_best)`.

The decorated variant in Table 7 (bottom row) adds back the multi-resolution + KAN output head; it is reproduced by dropping the `--single_head` and `--no_kan` flags from the recipe above. The other variants in Table 7 differ only in the additional CLI flag (`--mlp_groups 4`, `--mode_truncation disk`, `--spectral_envelope`, `--modes 16`, `--modes 20`).

H.4 Numerical reproducibility

We use a single seed (`torch.manual_seed(0)` + `numpy.random.seed(0)` + `generator.manual_seed(0)` on the DataLoader) for all reported point estimates; multi-seed results (Table 9) use seeds $\{0, 1\}$. We do not use deterministic cuDNN modes; cross-platform bit-identical reproduction is not guaranteed, but seed-0 reproductions on H100 + PyTorch 2.4 + CUDA 12.4 should agree to four decimal places on the headline test L^2.

I Broader impact

This work studies neural-operator surrogates for two-dimensional incompressible Navier–Stokes turbulence on synthetic, publicly-released benchmark data; no human-subject or personal data is used. The intended downstream uses are scientific (turbulence research, climate-model component surrogates, engineering fluid simulation) and edge-deployment of partial-differential-equation solvers (real-time control, interactive simulation tooling). Faster, lower-cost surrogates could incidentally accelerate dual-use applications such as aerospace or hydrodynamic design; this risk is diffuse and not specific to the architectural contributions reported here, which apply across all incompressible-flow regimes. We release code, data-preprocessing scripts, and trained checkpoints to support reproducibility and to make accuracy–cost trade-offs visible to deployment decisions (Section 9). The long-horizon-stability results in particular argue against the deployment of architectures that exhibit catastrophic divergence beyond the training horizon in safety-relevant rollouts.